Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

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Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

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A la meva família, que ha estat al meu costat en tot moment, donant-me l'empenta i el suport incondicional que necessitava, sobretot quan més em costava creure en mi mateix. I als meus amics, per ser la força que m'ha acompanyat dia a dia. Aquest projecte no marca només el final del grau, sinó el tancament d'una etapa vital molt important que m'ha canviat com a persona i l'inici d'un nou capítol.

A l'Hospital de Sant Pau i al Dimension Lab, per obrir-me les portes i ensenyar-me que el més bonic d'aquesta professió no són els algoritmes ni els números, sinó veure com allò que aprens pot servir per ajudar les persones i fer el bé. Sentir que aquest esforç pot tenir un impacte real i útil en la vida dels altres és la recompensa que dona sentit a tot el camí.

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Abstract

Coronary artery disease (CAD) assessment from coronary computed tomography angiography (CCTA) remains largely manual and time-consuming in clinical practice. At Hospital de la Santa Creu i Sant Pau, expert review of a single study can require between 30 minutes and two hours depending on case complexity. This bachelor's thesis presents an automated, end-to-end computational pipeline that transforms segmented coronary anatomy into quantitative stenosis metrics, CAD-RADS 2.0-oriented reporting support, and an interactive clinical visualization dashboard. The system is organized into four modular blocks: automated anatomy extraction, geometric stenosis quantification, segment labeling with CAD-RADS 2.0 scoring, and a Streamlit-based decision-support interface. Designed as a transparent in-house framework rather than a proprietary black box, the pipeline aims to reduce reporting burden, improve reproducibility, and assist patient prioritization while preserving final clinical authority. Validation combines ASOCA and MACS-18 cohorts with synthetic phantoms for controlled geometric verification.

Resum

L'avaluació de la malaltia coronària (MAC) a partir de la tomografia computada coronària (TCCC) continua sent, en la pràctica clínica, un procés manual i molt costós en temps. A l'Hospital de la Santa Creu i Sant Pau, la revisió experta d'un estudi pot requerir entre 30 minuts i dues hores segons la complexitat del cas. Aquest treball de fi de grau presenta un pipeline computacional automatitzat d'extrem a extrem que transforma l'anatomia coronària segmentada en mètriques quantitatives d'estenosi, suport orientat al CAD-RADS 2.0 i un tauler clínic interactiu de visualització. El sistema s'organitza en quatre blocs modulares: extracció automatitzada de l'anatomia, quantificació geomètrica de l'estenosi, etiquetatge segmentari amb puntuació CAD-RADS 2.0 i una interfície Streamlit de suport a la decisió. Dissenyat com a marc transparent i intern, el pipeline pretén reduir la càrrega de informes, millorar la reproduïbilitat i assistir la priorització de pacients sense substituir l'autoritat clínica final. La validació combina les cohorts ASOCA i MACS-18 amb fantasmes sintètics per a la verificació geomètrica controlada.

Resumen

La evaluación de la enfermedad coronaria (EC) a partir de la angiografía por tomografía computarizada coronaria (ATCC) sigue siendo, en la práctica clínica, un proceso manual y muy costoso en tiempo. En el Hospital de la Santa Creu i Sant Pau, la revisión experta de un estudio puede requerir entre 30 minutos y dos horas según la complejidad del caso. Este trabajo de fin de grado presenta un *pipeline* computacional automatizado de extremo a extremo que transforma la anatomía coronaria segmentada en métricas cuantitativas de estenosis, soporte orientado al CAD-RADS 2.0 y un panel clínico interactivo de visualización. El sistema se organiza en cuatro bloques modulares: extracción automatizada de la anatomía, cuantificación geométrica de la estenosis, etiquetado segmentario con puntuación CAD-RADS 2.0 e interfaz Streamlit de apoyo a la decisión. Diseñado como un marco transparente e interno, el *pipeline* pretende reducir la carga de informes, mejorar la reproducibilidad y asistir la priorización de pacientes sin sustituir la autoridad clínica final. La validación combina las cohortes ASOCA y MACS-18 con fantomas sintéticos para la verificación geométrica controlada.

Keywords

Coronary Artery Disease (CAD), Coronary Computed Tomography Angiography (CCTA), Automated Segmentation of Coronary Arteries (ASOCA), Coronary Artery Disease Reporting And Data System (CAD-RADS 2.0), Stenosis Quantification, Percentage Area Stenosis (%AS), Vascular Centerlines, Three-Dimensional Visualization, Clinical Decision Support, Automated Reporting, Patient Prioritization.

Preface

This document presents the bachelor's thesis completed during the 2025–2026 academic year within the Bachelor's Degree in Mathematical Engineering in Data Science at Universitat Pompeu Fabra, in collaboration with the Dimension Lab at Hospital de la Santa Creu i Sant Pau.

The main body of the thesis (Introduction through Conclusions) is written in English for consistency with the scientific literature in the field.

Use of Generative AI Disclosure

The Author, Adrià Cortés Cugat, discloses that the writing of this document "Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support" involved the use of Cursor AI and Google Gemini AI, for the purpose of: language review; text editing; technical writing support; LaTeX integration and formatting; software development assistance; code debugging and optimization; and pipeline documentation support. The Author warrants, however, that the contents of this thesis are a genuine human creation, as the results obtained from Cursor AI and Google Gemini AI have been duly reviewed and edited by Adrià Cortés Cugat. In accordance with university guidelines on the use of AI in assessable work, the technical and clinical substance of the main body, especially the Introduction, Methods, and Conclusions, has been written, validated, and revised by the Author with the support of the thesis director and supervisor.

Information source disclosure. The Author declares that all cited sources have been appropriately reviewed and validated by Adrià Cortés Cugat to support the claims made in this work, ensuring they are sufficiently contrasted and reliable.

Note on Figures and Data

The diagram figures that illustrate clinical and reporting workflows have been entirely designed and created by the Author. Where other figures build upon internal hospital materials, they are explicitly cited in the corresponding figure captions. Clinical cohort data and pipeline results are documented in the corresponding chapters and, where appropriate, in the supporting appendix.

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Chapter 1

INTRODUCTION

1.1. Clinical Context and Motivation

1.1.1. Anatomy of Coronary Arteries

The human heart is the central organ of the cardiovascular system and requires a continuous supply of oxygen and nutrients to function properly. This supply is provided by the coronary artery tree, a specialized vascular network that delivers blood directly to the heart muscle, known as the myocardium. The coronary arteries originate from the root of the aorta and surround the surface of the heart in a crown-like arrangement, from which the term *coronary artery* is derived [12]. As illustrated in Figure 1.1, the coronary arterial system is mainly composed of two vessels: the right coronary artery (RCA) and the left coronary artery (LCA).

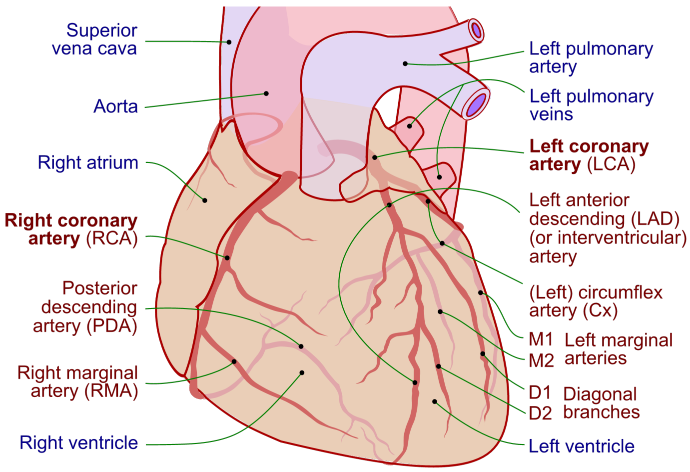

Figure 1.1: Anterior view of the human coronary tree; right coronary artery (RCA) and left coronary artery (LCA) are highlighted in red, with other cardiac structures in blue. Adapted from Lynch et al. [25].

The RCA primarily supplies blood to the right atrium and right ventricle. In contrast, the LCA quickly divides into two major branches: the left anterior descending (LAD) artery and the circumflex (LCx), which together supply the majority of the left ventricle and the interventricular septum [12]. From these primary vessels, a network of smaller branches extends deep into the myocardium to ensure regional blood delivery across the entire heart.

A critical characteristic of the coronary circulation is its limited collateral connectivity. Unlike other vascular systems in the human body, coronary arteries generally lack sufficient alternative pathways to redirect blood flow when an obstruction occurs. Consequently, any narrowing or blockage in a proximal arterial segment directly compromises blood flow to all downstream regions supplied by that vessel [30].

Additionally, coronary anatomy presents significant inter-patient variability in terms of vessel size, branching structure, and coronary dominance. Dominance is defined by which artery gives rise to the posterior descending artery (PDA) and can be classified as right-dominant, left-dominant, or codominant. Right-dominant circulation, in which the PDA originates from the RCA, is the most common configuration, followed by left dominance and codominance [34]. Understanding this complex anatomical layout is fundamental, as it provides the basis for locating lesions, quantifying stenosis severity, and stratifying risk in coronary artery disease (CAD) diagnosis.

1.1.2. Coronary Artery Disease

Coronary artery disease (CAD) remains one of the leading causes of morbidity and mortality worldwide [33]. It is characterized by the progressive accumulation of atherosclerotic plaque within the walls of the coronary arteries, leading to a narrowing of the vessel lumen known as stenosis. This narrowing restricts blood flow to the myocardium and, if left untreated, can result in myocardial ischemia or infarction [23, 26]. Figure 1.2 illustrates this pathological mechanism, contrasting a normal artery with one obstructed by plaque buildup.

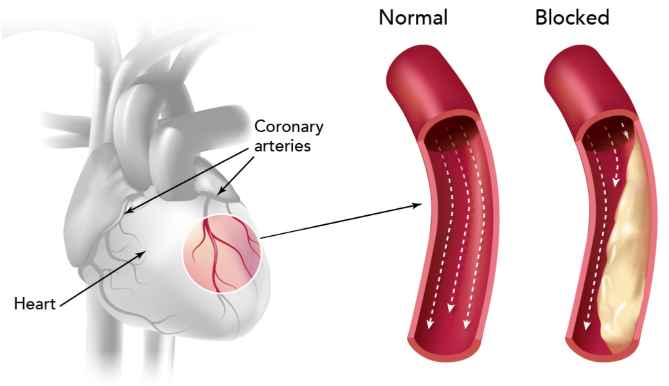

Figure 1.2: Coronary arteries narrowed by atherosclerotic plaque in coronary artery disease (CAD). Adapted from Kaiser Permanente [20].

Early and accurate diagnosis is essential for effective risk stratification and patient management. In current clinical practice, coronary computed tomography angiography (CCTA) is the standard non-invasive imaging technique for CAD evaluation [23]. CCTA provides detailed 3D reconstructions of the coronary tree, allowing clinicians to detect plaques, analyze their spatial location, and estimate the degree of luminal narrowing.

However, despite the high resolution of CCTA, the diagnostic workflow remains highly time-consuming and heavily dependent on manual and visual assessment. For instance, at Hospital de la Santa Creu i Sant Pau, reporting non-pathological or low-complexity cases requires approximately 30 minutes, while complex evaluations can take up to 2 hours [1]. This significant clinical burden drives the need for automated, data-driven tools capable of quantifying stenosis efficiently and supporting clinical decision-making, while fully preserving expert interpretability and clinical control.

1.1.3. Stenosis Quantification: A Key Indicator of CAD Severity

Among the various indicators used to assess coronary artery disease, stenosis severity plays a central role in clinical diagnosis and risk stratification. Stenosis represents the reduction of the vessel lumen caused by atherosclerotic plaque accumulation, a condition directly associated with an increased risk of myocardial ischemia and adverse cardiac events [15, 23]. In clinical practice, this severity is quantified using geometric measurements such as the minimal lumen area (MLA) or percentage diameter stenosis (%DS). These metrics evaluate the degree of narrowing by comparing the most constricted point of a lesion with a reference baseline, which is typically estimated from adjacent, relatively healthy vessel segments [15, 17].

Although these definitions are conceptually simple, accurate stenosis quantification presents significant clinical challenges. Coronary arteries exhibit complex, tortuous geometries, and diffuse atherosclerosis often makes it difficult to identify truly healthy reference regions. In addition, limitations in image resolution, segmentation errors, and noise in CCTA-derived data can highly influence measurements, leading to variability in stenosis estimates [17, 21].

In routine clinical practice, stenosis assessment is often performed visually or using semi-automatic tools provided by proprietary software [1, 13]. Although these methods are widely used, they depend heavily on the clinician's interpretation and can vary between observers, which reduces consistency and reproducibility [17].

Consequently, there is a strong clinical need to develop robust, transparent, and reproducible quantification methods. Automating these geometric measurements has the potential to standardize stenosis evaluation, reduce reporting time, and provide reliable quantitative data essential for downstream clinical tasks, such as CAD-RADS 2.0 assignment and patient prioritization.

1.1.4. CAD-RADS 2.0 and the Need for Decision Support

To standardize the reporting of coronary disease and facilitate clinical communication, the Coronary Artery Disease-Reporting and Data System (CAD-RADS 2.0) was established [13]. This structured framework categorizes patients into distinct risk scores ranging from 0 to 5, primarily based on the most severe stenosis measured via CCTA. Each categorical score reflects the overall disease burden and provides specific clinical recommendations, offering a standardized pathway for subsequent patient management [13].

While CAD-RADS 2.0 significantly improves reporting consistency across institutions, its assignment still relies heavily on the manual interpretation of stenosis severity across multiple vessels and segments. This process requires clinicians to continuously synthesize quantitative geometric measurements with complex anatomical context. Consequently, evaluating these scans becomes highly cognitively demanding, particularly in time-constrained clinical environments [1]. In practice, manual scoring remains susceptible to inter-observer variability, especially in borderline cases or in patients presenting with multiple moderate lesions. Such discrepancies can lead to inconsistent diagnostic classifications, ultimately impacting downstream clinical decisions, further testing, and patient prioritization [1, 17].

To address these limitations, there is a growing clinical imperative to develop automated decision support tools. These solutions can span from rule-based algorithms mapping quantitative stenosis measurements to machine learning approaches that leverage structured imaging features. Crucially, these computational tools aim to provide objective, reproducible, and transparent suggestions to assist radiologists and cardiologists. By streamlining the CAD-RADS 2.0 assignment process, these systems reduce reporting times and variability while strictly preserving expert clinical judgment and final decision authority.

1.2. Clinical Automation Challenges at Hospital de la Santa Creu i Sant Pau

1.2.1. Current CAD Diagnostic Workflow and Associated Limitations

The current diagnostic workflow for Coronary Artery Disease (CAD) at Hospital de la Santa Creu i Sant Pau involves a complex and coordinated series of steps, managed primarily by the specialized cardiac imaging unit (Unitat d'Imatge i Funció Cardíaca, UIFC). As illustrated in Figure 1.3, the clinical pathway begins when a patient presenting with symptoms such as chest pain is scheduled for a Coronary Computed Tomography Angiography (CCTA). Once the scan is acquired, the images are stored in the hospital's *Picture Archiving and Communication System* (PACS) for further evaluation.

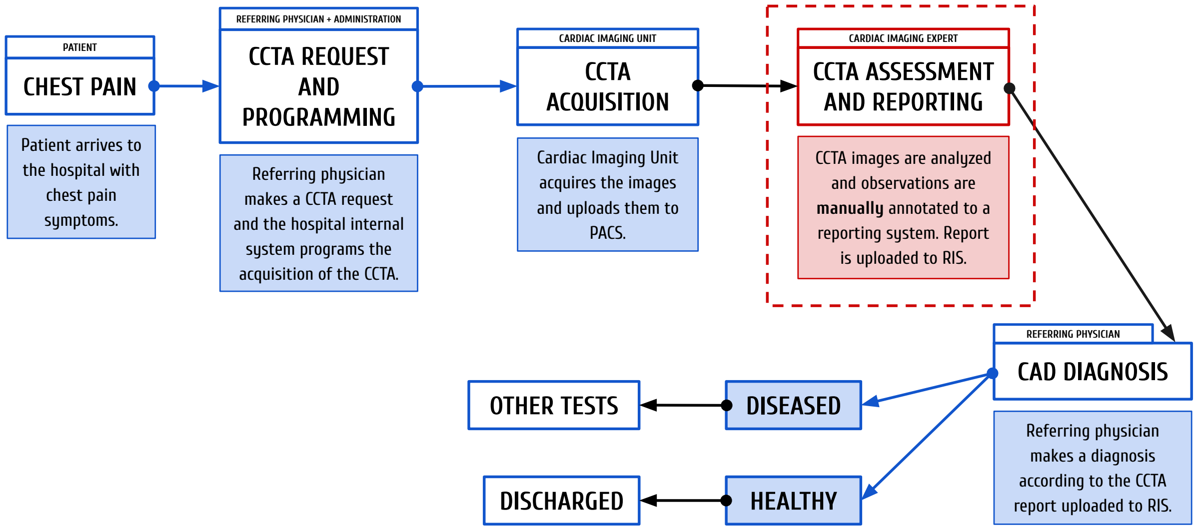

Figure 1.3: Current CAD diagnostic workflow at Hospital de Sant Pau, spanning coronary computed tomography angiography (CCTA) acquisition, picture archiving and communication system (PACS) storage, and radiology information system (RIS) reporting. The primary operational bottleneck (image assessment and reporting) is highlighted in red. Adapted from Ferrer Beltran [14].

The core of this diagnostic pipeline relies heavily on the visual and manual analysis of these scans by expert cardiologists and radiologists to extract clinical metrics. These findings are then compiled into a clinical report, which the referring clinician evaluates to determine the appropriate subsequent medical actions, ranging from discharging a healthy patient to programming further invasive tests. While the overall infrastructure is well-established, the central phase of image analysis and reporting (highlighted in red in Figure 1.3) represents a critical operational bottleneck.

1.2.2. The Bottleneck: Manual Image Analysis and Reporting

Zooming into the specific image analysis and reporting phase (Figure 1.3), the limitations of the current clinical pathway become evident. Currently, clinicians rely on commercial software solutions, such as *syngo.via*, to manually inspect the coronary arteries, differentiate segments, and calculate stenosis severity. Although these platforms offer semi-automated tools, the process remains highly dependent on the clinician's expertise to visually verify the vessels, identify bifurcations, and assess the plaque burden. Depending on the anatomical complexity and image quality, analyzing a single patient can consume around 30 minutes for healthy cases and 90 minutes for complex cases [1].

Once the visual assessment is complete, the extracted metrics are manually entered into an internal custom-built platform called *Filemaker*, which generates a template-based preliminary report. The specialist must then review, correct, and finalize this text before it is integrated into the *Radiology Information System* (RIS).

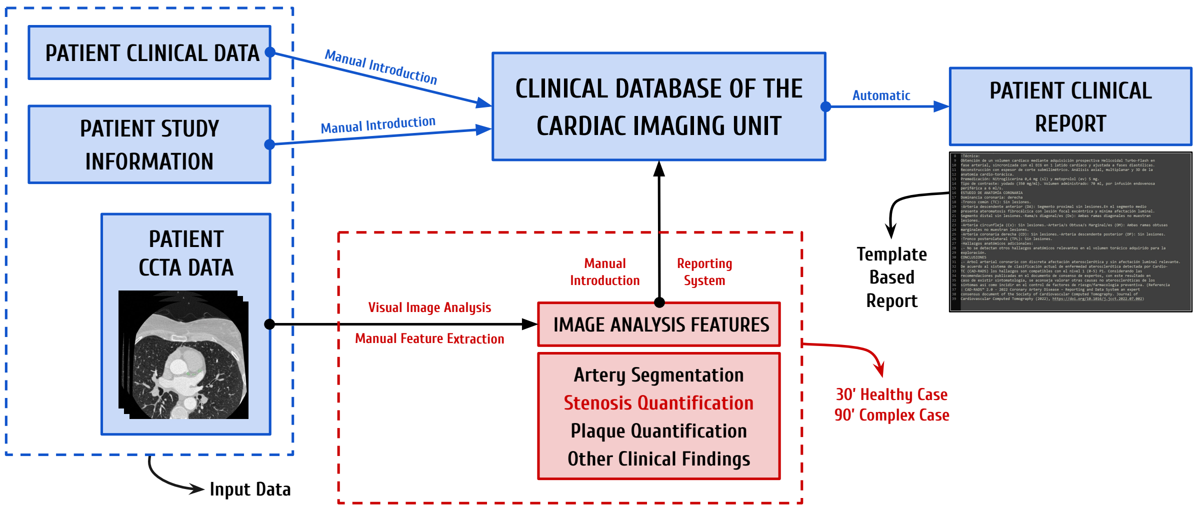

Figure 1.4: Detailed representation of the CCTA image analysis and reporting workflow at Hospital de Sant Pau. Adapted from Acebes Pinilla [1].

This heavy reliance on manual quantification and data entry presents significant operational challenges. Consequently, highly specialized experts spend considerable time performing repetitive, manual tasks on low-risk patients, rather than focusing their expertise on critical cases. This lack of an automated, prioritized workflow delays the diagnosis of severe cases and highlights a clear necessity for robust and automated solutions.

1.3. State of the Art: Automation of CAD Diagnosis and Visualization

The integration of Artificial Intelligence (AI) and advanced computer vision in cardiac imaging has seen exponential growth over the past decade, driven by the need to optimize clinical workflows [24]. Deep learning algorithms, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in medical image analysis, encouraging a move from manual CCTA assessment to automated systems [31]. However, significant limitations remain regarding their practical implementation. For these new techniques to succeed, they must be practical for hospitals, integrate smoothly into current systems, and be easily adopted by the clinical staff [22].

The prerequisite first step in any automated CAD assessment pipeline is the accurate segmentation and anatomical labeling of the coronary artery tree; before any downstream diagnostic metric can be calculated, the vessels must be correctly isolated and identified. Recent advancements have leveraged deep learning architectures, such as 3D U-Nets, to extract the vessel lumen from CCTA images [9], while graph-based models like Graph Convolutional Networks (GCNs) are increasingly used to ensure topological consistency when labeling segments according to standard clinical guidelines [35]. Within Hospital de la Santa Creu i Sant Pau, preliminary research has explored these initial segmentation and labeling tasks using transformer-based models [10, 29].

Following successful segmentation, extracting the precise geometric properties of the vessels is essential. The Vascular Modeling Toolkit (VMTK) stands out as one of the most prominent open-source libraries for medical geometry data extraction [19]. Specifically, algorithms implemented within VMTK are widely utilized to extract accurate vascular centerlines and compute cross-sectional areas [19, 27]. These metrics provide the essential topological framework required to perform continuous Multi-Planar Reformations and subsequently evaluate vessel narrowing.

Once the 3D geometry of the coronary tree is established, the focus shifts to quantifying the disease burden. Current stenosis quantification techniques range from traditional intensity-based thresholding (which often struggles with severe calcifications) to modern machine learning models that predict stenosis severity directly from image features [18, 36]. Alternatively, geometric, non-ML approaches offer highly interpretable and robust methods by calculating the percentage of diameter or area stenosis based purely on the physical vessel profile, comparing the minimal lumen area against an interpolated healthy reference [17]. Recognizing the clinical value of algorithmic transparency, ongoing research at Hospital de la Santa Creu i Sant Pau has actively explored these purely geometric techniques to quantify stenosis, establishing a reliable, explainable foundation for disease assessment [8].

Beyond quantification, the effective visualization of these complex 3D metrics remains a critical challenge in the state of the art [6]. In the current clinical routine at Hospital de Sant Pau, specialists must manually navigate through cross-sectional slices and use standard viewing software to visually synthesize the data, assess the lesions, and manually fill out structured reports [1]. This cognitive load highlights a significant gap: the lack of dedicated, interactive visualization tools that bridge the gap between automated metrics and clinical reality. Modern research and clinical consensus emphasize that integrating 3D anatomical models alongside continuous luminal profiles and real medical images (such as Curved Multi-Planar Reformations) into a unified graphical interface is essential. Such integration allows clinicians to continuously validate automated findings against the ground-truth CCTA, drastically reducing the time and effort required to verify stenosis locations and seamlessly generate diagnostic reports [6].

Several commercial software solutions have emerged to address these overarching needs, including platforms like *Cleerly* [11] and *Artrya Salix* [3], which use AI to analyze arterial plaque and assess ischemia risk from CCTA images. While these robust dashboards offer highly desirable visualization components, they often function as proprietary “black boxes”. This lack of algorithmic transparency significantly hinders explainability, a feature strongly demanded by medical experts to verify how a specific CAD-RADS 2.0 category, plaque distribution, or stenosis percentage was computed. Finally, these standalone commercial platforms frequently struggle to integrate well into the highly specific IT infrastructures and patient prioritization workflows of public hospitals [22].

1.4. Project Scope and Contributions

To address the clinical challenges outlined in Section 1.2, and to overcome the limitations of commercial tools discussed in Section 1.3, this project contributes directly to the comprehensive, transparent, and in-house AI-enabled diagnostic framework developed by the Dimension Lab at Hospital de la Santa Creu i Sant Pau [1]. While previous academic efforts have tackled earlier stages of this pipeline, such as patient prioritization [14] and coronary artery segmentation and labeling [10], a critical gap remains in translating raw segmented geometries into actionable clinical insights.

Therefore, the scope of this bachelor’s thesis focuses on automating the geometric quantification of coronary stenosis, assigning the associated CAD-RADS 2.0 category through rule-based scoring, and integrating these findings into an interactive clinical dashboard. To realise this goal, this project develops a custom Python-based computational pipeline that implements the automated image analysis and feature extraction stages outlined in Figure 1.5. This end-to-end framework bridges the gap between raw imaging data and the clinical reporting workflow.

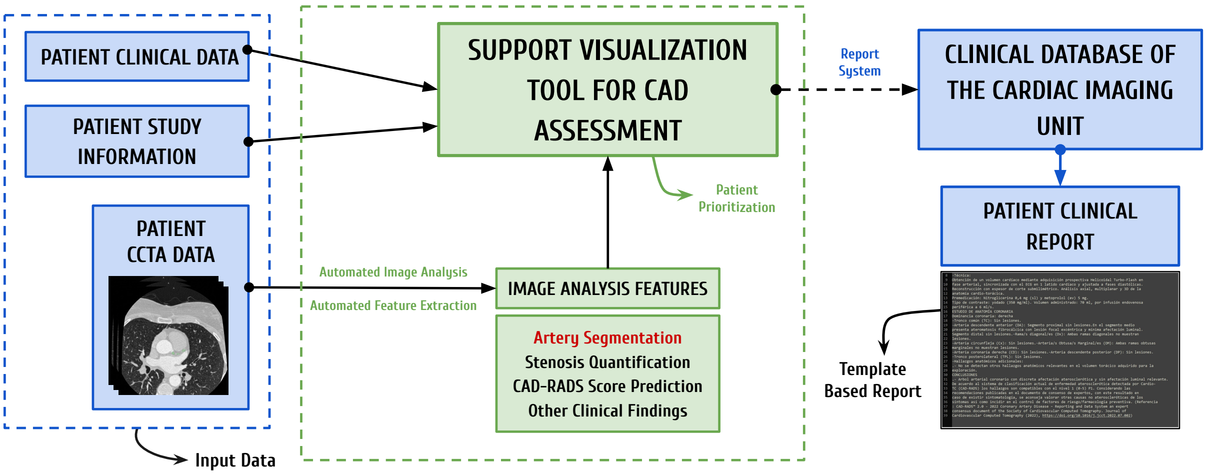

Figure 1.5: Contextual overview of the proposed automated CAD diagnostic framework at Hospital de la Santa Creu i Sant Pau, integrating CCTA-based analysis and Coronary Artery Disease Reporting and Data System (CAD-RADS 2.0) scoring. The contributions of this thesis (green dashed box) bridge the gap between existing hospital data systems (blue) and the final clinical report. Adapted from Acebes Pinilla [1].

By seamlessly connecting raw image analysis with the hospital's reporting systems, this project aims to mitigate operational bottlenecks and reduce overall diagnostic times. However, the solution is intended solely as a clinical decision-support tool, not as a replacement for medical expertise or detailed diagnostic software. Designed as a rapid initial assessment interface, it presents an immediate, high-level overview of each patient's critical metrics to help specialists evaluate prioritized patient lists and identify cases requiring in-depth analysis. In this way, the automated outputs speed up the workflow, but doctors always retain responsibility for the final diagnosis and clinical validation.

1.5. Objectives

The primary objective of this thesis is to design and develop an automated, end-to-end pipeline that successfully bridges the current gap in the CAD diagnostic workflow. Importantly, this project does not seek to maximize the precision of individual algorithmic components; rather, the proposed functional workflow is, in itself, the core product of this work.

To achieve this overarching goal, the project addresses a sequence of intermediate objectives. First, it requires deep contextualization within the complex clinical environment to evaluate prior research conducted at the hospital and design a unified, clinically applicable framework. Building upon this foundation, the project aims to implement computational geometry algorithms to accurately extract cross-sectional areas and calculate the percentage of area stenosis from 3D models. Subsequently, a standardized logic is developed to translate these geometric metrics into a reliable, rule-based CAD-RADS 2.0 category assignment.

To aggregate these findings and provide cardiologists with interpretable metrics, an interactive clinical dashboard is built. Functioning as a rapid-access pop-up interface, this Support Visualization Tool gives clinicians an immediate, visual summary of the patient's coronary status, ultimately reducing medical workload, optimizing case triaging, and significantly decreasing overall assessment time. Finally, a technical validation is conducted to verify the reliability of the extracted metrics and ensure the overall stability of the complete framework.

Chapter 2

METHODS

2.1. System Overview and Core Automated Workflow

This section formalizes the modular Python pipeline introduced in Section 1.4, translating the clinical decision-support framework into an executable computational workflow. For each patient identifier, a pre-segmented coronary lumen mask is processed sequentially into quantitative stenosis metrics, a CAD-RADS 2.0-oriented report, and an interactive visualization session.

The workflow comprises four automated blocks (Sections 2.3 to 2.6) executed as a linear chain, with each stage consuming the artifacts produced by its predecessor and no manual intervention between steps. A unified patient identifier routes ASOCA, MACS-18, and synthetic cohorts through the same implementations, while runtimes and extraction metrics are logged for Section 2.7.

2.1.1. End-to-End Data Flow

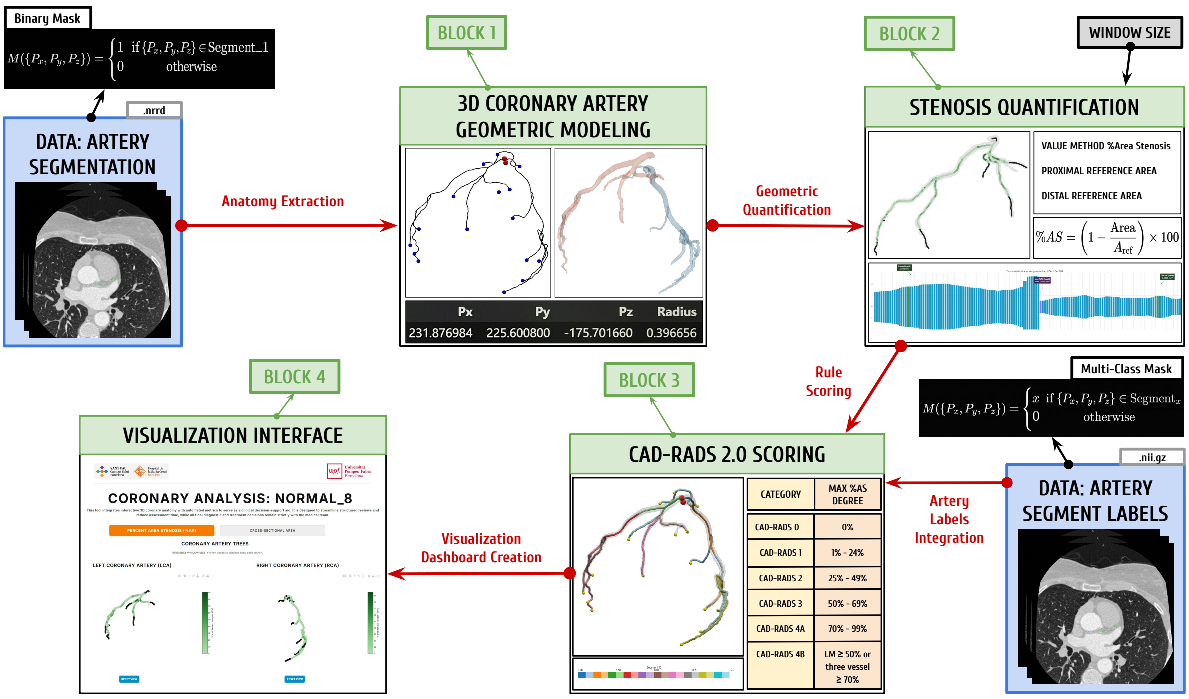

Figure 2.1: End-to-end per-patient workflow: lumen mask, percentage area stenosis (%AS), and CAD-RADS 2.0 reporting. Green boxes denote workflow blocks; blue boxes denote input data; arrows show sequential data flow.

Figure 2.1 summarizes the four-block architecture. The workflow assumes as input a pre-segmented coronary lumen mask and, when available, anatomical segment labels from prior segmentation and labeling work at the hospital [10].

Table 2.1: Overview of the four pipeline blocks: processing role, primary inputs, and principal outputs per patient case. Block 2 yields %AS profiles; Block 3 produces the CAD-RADS 2.0 report.

Block	**Primary input(s)**	**Principal output(s)**
1 Anatomy Extraction	Binary lumen mask	Centerline geometry, branch paths, and vessel surface models
2 Stenosis Quantification	Centerline and surface representations from Block 1	Cross-sectional area profiles, percentage area stenosis (%AS), and merged branch-level data
3 Labeling & CAD-RADS 2.0	Geometric and stenosis tables	Segment stenosis summary and patient CAD-RADS 2.0 report
4 Visualization	Aggregated metrics and geometry from Blocks 1–3	Interactive clinical decision-support dashboard

To ensure explainability, the system retains intermediate mesh and tabular representations for potential clinical audits. The clinical cohorts utilized, along with the methodological depth of each processing block, are detailed below.

2.2. ASOCA and MACS-18: Coronary Segmentation and Labeling Data

In medical image analysis, *segmentation* denotes the partition of a three-dimensional scan into meaningful regions. For each voxel in the volume, an expert or algorithm decides whether it belongs to a structure of interest, such as the blood-filled interior (*lumen*) of the coronary arteries, rather than to background tissue, myocardium, or surrounding anatomy. The result is not a single measurement but a spatial map that localizes the vessel tree within the patient-specific CCTA grid. The ASOCA and MACS-18 resources provide this map, together with companion anatomical label volumes for a shared cohort of 40 CCTA cases (20 normal and 20 diseased anatomy), which are detailed further in Section 2.2.1. These data served as the reference on which Blocks 1–4 were designed and implemented. Once the pipeline was operational, the full cohort was processed to assess end-to-end behavior. The experimental protocol and result analysis are reported in Section 2.7.

2.2.1. Datasets: ASOCA and MACS-18

The ASOCA (*Automated Segmentation of Coronary Arteries*) challenge established one of the first large, standardized benchmarks for fully automatic segmentation of complete normal and diseased coronary trees from CCTA [16]. The resource addresses the difficulty of imaging small, moving vessels. ASOCA is publicly available and provided the original lumen masks for the 40-patient cohort. In this project, those masks served as the working reference during pipeline development and remain the default segmentation supplied to Block 1.

MACS-18 (*Multiclass Anatomical Coronary Segmentation*) is a re-segmentation of the same 40 ASOCA patient volumes, currently in development at Hospital de la Santa Creu i Sant Pau and supervised by César Acebes Pinilla [1, 2]. Unlike ASOCA, this resource is not yet publicly available. Widely used public datasets have helped train automatic segmentation methods, but their lumen masks are often evaluated mainly by how well voxels match a reference, not by whether the vessel tree forms a continuous, anatomically plausible path [2]. When the lumen mask contains breaks or fails to follow the full length of the branches, extracting reliable centerlines and quantifying vessel geometry in Blocks 1 and 2 becomes more difficult. Expert manual refinements in MACS-18 target continuous lumen connectivity from the aortic root to the most distal visible branches, yielding more anatomically faithful masks without altering the underlying imaging data, patient identity, or labeling scheme. In this project, the same AHA segment label volumes are used for both ASOCA and MACS-18 cases.

2.2.2. Segmentation and Label Volumes

The previously described clinical datasets (ASOCA and MACS-18) provide two aligned 3D grids per patient derived from source CCTA images; extracting those segmentations is outside this project's scope, so the pipeline relies on pre-segmented inputs. These grids are spatially registered to the CCTA coordinate system, preserving physical scale and anatomical orientation.

The *segmentation annotation* is stored as a voxel grid in NRRD format. Each voxel indicates whether it belongs to the coronary lumen or to background tissue. The pipeline treats this volume as a binary mask.

The *segment label volume* is a separate grid, stored as a compressed NIfTI file, aligned with the segmentation. Instead of a binary lumen flag, each foreground voxel stores an integer class identifier that maps to a standardized coronary segment under the AHA 17-segment model [4]. Background voxels are assigned zero.

2.3. Block 1: 3D Artery Anatomy Extraction

Block 1 converts the binary lumen mask into 3D vessel surfaces and centerlines, recovering both global RCA and LCA geometries and individual ostium-to-tip branch paths that provide the anatomical framework for stenosis quantification in Block 2.

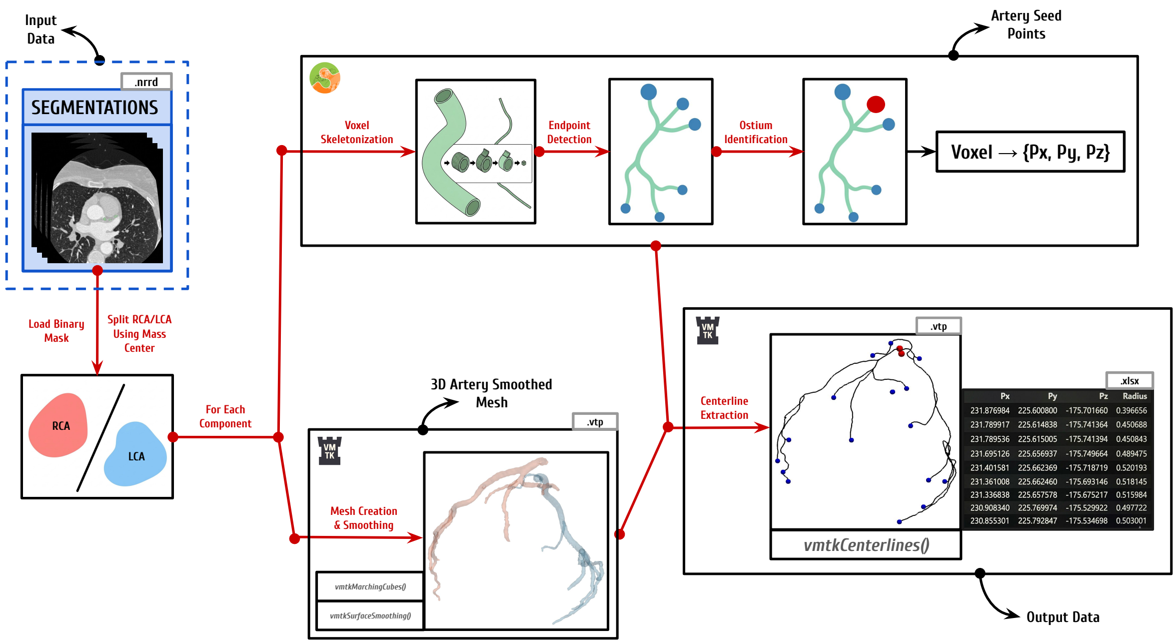

Figure 2.2: Block 1 hybrid workflow: separation of right coronary artery (RCA) and left coronary artery (LCA) trees; automated per-tree seed placement; centerline geometry extraction; branch packaging.

As illustrated in Figure 2.2, this block implements a hybrid approach. Instead of relying on a single tool, it combines automated voxel analysis with the surface and centerline extraction algorithms of the Vascular Modeling Toolkit (VMTK) [19]. This design completely removes the need for manual seed-point selection while maintaining the precise 3D shape of the smoothed vessels.

Phase 1: Spatial RCA/LCA Separation

The process begins by reading the binary lumen mask using `vmtkImageReader` to preserve voxel spacing and physical origin. To process the Right Coronary Artery (RCA) and Left Coronary Artery (LCA) independently, `scipy.ndimage.label` is applied to isolate the two largest connected components. In order to correctly classify these isolated components into the main RCA and LCA trees, the algorithm calculates their spatial center of mass. This metric represents the average 3D coordinate of all voxels within each object. By sorting these center points along the patient's spatial x-axis, the system follows standard anatomical conventions: the rightmost component is classified as the RCA, and the leftmost as the LCA. This splitting step produces two independent binary sub-volumes. Processing them separately is logically necessary before geometric extraction, as VMTK algorithms require a single starting source per connected surface.

Phase 2: Per-Artery Surface Reconstruction and Automated Seeding

The two independent binary sub-volumes outputted by Phase 1 serve as the direct input for the surface reconstruction. For each isolated tree, a 3D surface mesh is generated using `vmtkMarchingCubes`. This function creates a continuous geometric boundary formed by connected triangles by interpolating the exact boundary between the foreground lumen voxels and the background. The raw mesh is then refined via `vmtkSurfaceSmoothing` using a pass-band filter. In this context, a pass-band filter works by separating high-frequency noise from the actual anatomical structure. It removes the rough, blocky edges caused by the original voxels and this creates a clean, mathematically stable surface for the next calculations without altering the true size of the artery.

The next step requires placing seed points, which are the explicit starting and ending spatial coordinates required by VMTK to trace the centerlines. To fully automate this placement and eliminate manual user interaction, the system analyzes the overall 3D shape of the voxel mask to automatically find key structural landmarks. The algorithm to obtain these seed points works by iteratively "peeling" away the outer layers of the voxel grid (a process known as skeletonization) and was leveraged from prior work [10]. The volume is thinned using `skimage.morphology.skeletonize`, reducing the complex 3D shape down to a one-voxel-wide medial axis. Topological endpoints are then easily identified as voxels that have exactly one neighboring voxel (a node degree of one).

The ostium, where the artery leaves the aorta and centerline tracing must begin, is selected among the skeleton endpoints using the AHA label volume (Section 2.2.2). Under the AHA 17-segment model, segment 1 identifies the proximal RCA and segment 5 the proximal LCA [4]; an endpoint inside either region is designated as the ostium point. If none qualifies, the widest endpoint by maximum inscribed sphere radius is used as a fallback.

All remaining endpoints are designated as distal targets. Because these endpoints are initially identified in a discrete voxel grid (i, j, k), they must be mapped to the continuous physical space (x, y, z) to be compatible with the mesh. This is achieved using the image origin (O_x, O_y, O_z) and the voxel spacing (S_x, S_y, S_z) through the following spatial transformation:

$$x = O_x + i\,S_x, \quad y = O_y + j\,S_y, \quad z = O_z + k\,S_z. \tag{2.1}$$

Once mapped, these physical coordinates are projected onto the nearest vertex of the smoothed mesh to ensure the seed points lie exactly on the mathematical manifold.

Phase 3: Centerline Extraction, Topology Labeling, and Branch Packaging

With the ostium and distal seeds fixed on the smoothed surface mesh, `vmtkCenterlines` [19, 27] is executed to trace the coronary centerlines. For each ostium-to-distal pair, the function extracts the most centered route through the vessel medial axis and resamples it into a regular ordered sequence of 3D points. These 1D curves map the branching coronary structure. At every resampled point, VMTK records the maximum inscribed sphere radius under `MaximumInscribedSphereRadius` as an initial estimate of local vessel width. The VMTK centerline geometry algorithm underlying this step is described in greater detail in Section A.1.

Once the centerlines are generated, a labeling step assigns a specific topological point type to every coordinate based on its structural position. To guarantee anatomical validity for downstream stenosis quantification, the centerlines are partitioned into continuous branch paths running directly from the ostium to each distal endpoint. Geometrical artifacts (unintended or anomalous extraction traces) are filtered out by discarding short branches with fewer than 20 points or paths that fail to terminate at a valid distal endpoint. The 20-point minimum was set empirically during pipeline development to remove spurious short traces while retaining the main coronary branches.

Finally, the extracted data is packaged into a structured format. The spatial coordinates (P_x, P_y, P_z), local radius, assigned point type, and corresponding AHA segment identifiers are stored as ordered tables for both the individual branches and the global coronary tree. The generated surface and centerline meshes are also saved in `.vtp` format for visual rendering. Concluding Block 1, the RCA and LCA tables are concatenated into a unified patient record, providing the exact geometric dataset required for Block 2.

2.4. Block 2: Geometric Stenosis Quantification

Block 2 transforms the Block 1 centerline and surface representations into percentage area stenosis (%AS) profiles by measuring cross-sectional lumen area and comparing each point against a locally interpolated healthy reference.

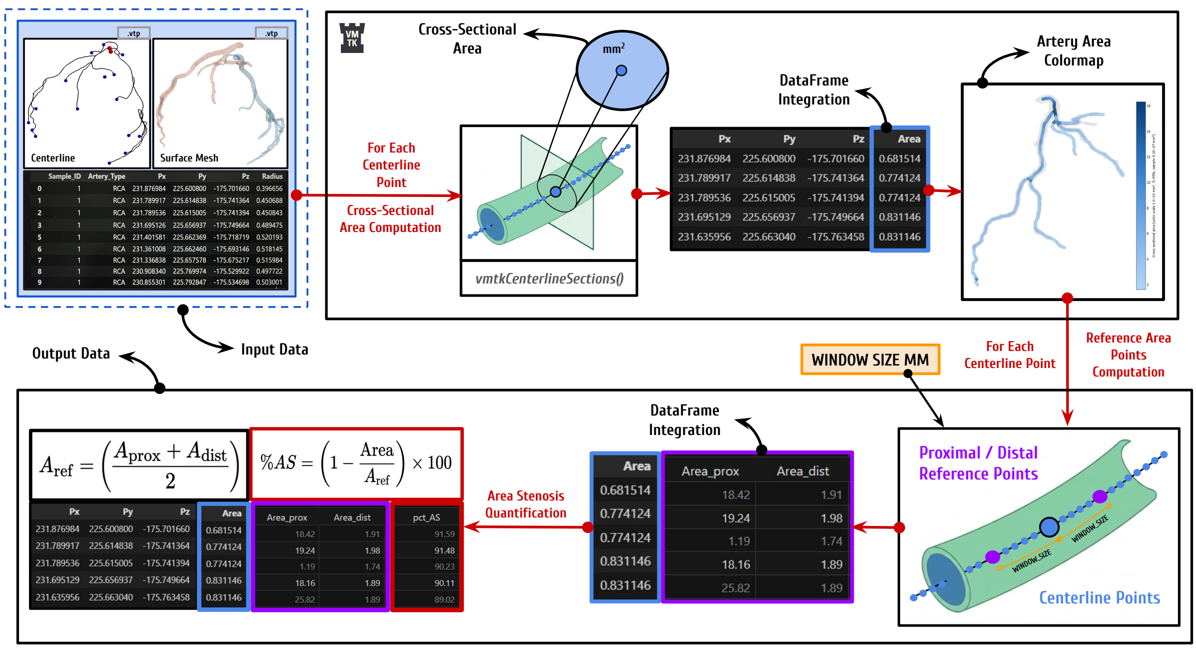

Figure 2.3: Block 2 workflow: orthogonal cross-sectional area extraction on each artery centerline and surface mesh, tabular integration, proximal/distal reference estimation, and %AS quantification.

As illustrated in Figure 2.3, Block 2 processes each available artery (RCA and LCA) independently. For every tree, it consumes the paired centerline and smoothed surface meshes produced in Section 2.3.

The computational logic for stenosis quantification and the methodology used to identify local healthy reference values build upon prior internal research by Burrull [8], originally grounded in the standard percentage diameter stenosis (%DS) framework described by Hideo-Kajita et al. [17]. Rather than reproducing diameter-based scores directly, this pipeline adapts those reference principles to cross-sectional *area* and percentage area stenosis (%AS).

In coronary arteries, atherosclerotic plaque frequently develops eccentrically rather than as a perfect concentric ring. Relying solely on one-dimensional diameter estimates, such as the Maximum Inscribed Sphere Radius (MISR), can therefore misrepresent luminal narrowing. In an elliptical or irregularly narrowed vessel, the inscribed sphere captures only the narrowest physical constraint and ignores the lateral space through which blood may still flow.

Conversely, the cross-sectional area captures the absolute reduction of the luminal opening regardless of cross-sectional shape. This yields a more robust representation of flow restriction and a geometric metric that correlates more closely with hemodynamically significant stenosis than relative diameter-based estimates [28]. The following phases mirror the diagram, detailing execution from localized area computation through patient-level aggregation.

Phase 1: Cross-Sectional Area Extraction

The process begins by lightly smoothing the global centerline and resampling it to establish a uniform spatial resolution. The polyline is interpolated so that the physical distance between consecutive discrete points is standardized to 0.1 mm by default. While larger intervals could reduce computational load, this spacing is deliberately chosen to capture high-fidelity geometric detail and subtle anatomical variations along the vessel. For exceptionally long arterial trees, this spacing is adaptively increased to ensure the total number of evaluated points does not exceed a maximum limit of 12,000 per component.

At every resampled centerline point, `vmtkCenterlineSections` [19, 27] computes the cross-sectional lumen area at that specific point. To do this, the algorithm finds the local direction (tangent) of the centerline and creates a flat, perpendicular cutting plane. It then slices the 3D smoothed surface mesh (from Block 1) with this plane to extract the exact 2D boundary of the vessel lumen, calculating its enclosed area in mm^2. Any incomplete or open slices are automatically discarded.

Highly tortuous or irregular geometries can occasionally cause the native VMTK sectioning step [19] to fail, typically as an internal kernel error. To guarantee a complete, continuous area profile, a robust fallback replicates the same orthogonal slicing logic with Visualization Toolkit (VTK) operators when a failure is detected. At each evaluation point along the centerline, a `vtkPlane` (oriented perpendicularly to the centerline) combined with a `vtkCutter` slices the 3D surface to extract the cross-sectional lumen contour and compute its enclosed area in mm^2 using a standard polygon formula. This ensures accurate cross-sectional areas even in the most challenging anatomical segments.

Phase 2: Tabular Area Integration and Branch Mapping

To perform localized stenosis quantification, the cross-sectional areas computed across the global coronary tree must be accurately mapped to the individual ostium-to-endpoint branch dataframes generated in Block 1.

For every spatial coordinate (P_x, P_y, P_z) in a branch table, if the sequence of centerline points aligns perfectly with the global area output, the value is assigned via direct row matching. If the points do not align exactly, the system uses a `scipy.spatial.cKDTree` nearest-neighbor query to find the closest evaluated station in 3D space and map its computed area to the specific branch point.

Ultimately, this mapping step propagates a continuous `Area` metric to every discrete coordinate across all branch paths. It seamlessly merges the newly extracted geometry with the anatomical information inherited from Block 1, preparing the exact dataset required for the next baseline reference calculations.

Phase 3: Geodesic Distance and Reference Area Estimation

Stenosis severity is evaluated independently along each individual branch path rather than on the global coronary tree dataset. This isolation is critical: it ensures that spatial evaluation strictly follows the natural, antegrade blood flow from the ostium to the distal tip. Evaluating the tree globally could inadvertently select reference points that are spatially close in Euclidean space but belong to entirely different, non-connected vessels.

To measure distance along the tortuous 3D vessel, consecutive centerline points are connected by straight segments, and the cumulative sum of these segment lengths defines a one-dimensional geodesic distance (g_d). This g_d value is then integrated into the branch dataframe for every discrete coordinate.

To evaluate stenosis at any specific point, the algorithm requires a healthy reference area for comparison. The system first identifies proximal and distal centerline points located at geodesic positions $g_d - W$ and $g_d + W$, respectively. The reference half-window W is a crucial hyperparameter in the pipeline. Based on the geometric stenosis framework introduced by Burrull [8] and standard angiographic principles described by Hideo-Kajita et al. [17], W is set to 10 mm in this project. However, this 10 mm threshold is an empirical approximation: there is no universally optimal value, as the ideal reference distance depends on patient-specific anatomy and individual lesion length. Ideally, to maximize clinical accuracy, the system would allow a physician to manually select or adjust proximal and distal reference points based on visual anatomical assessment. Because the computed stenosis varies significantly with the automated window size, this introduces an important methodological limitation that is discussed later.

Once the two reference points are identified along the same branch, their corresponding mapped areas (A_{prox} and A_{dist}) are retrieved. The healthy reference area (A_{ref}) at the evaluation point is then calculated as the arithmetic mean of these bilateral values:

$$A_{\text{ref}} = \frac{A_{\text{prox}} + A_{\text{dist}}}{2}. \tag{2.2}$$

Because this formula requires both proximal and distal context, percentage area stenosis cannot be calculated for points at the very extremities of branch paths. Consequently, any evaluation point lying within distance W from the ostium or the distal tip is excluded, as it would possess only a one-sided reference. Alongside sensitivity to window size, additional anatomical challenges of this automated baseline (such as unpredictable behavior near major bifurcations) are addressed in Chapter 4.

Phase 4: Percentage Area Stenosis and Patient-Level Aggregation

For each qualifying centerline point, the pipeline quantifies stenosis severity. The percentage area stenosis (%AS) formula, adapted from Burrull [8], compares the local cross-sectional lumen area (A) directly against its corresponding healthy reference (A_{ref}):

$$\%\text{AS} = \left(1 - \frac{A}{A_{\text{ref}}}\right) \times 100. \tag{2.3}$$

Next, the separate branch tables are merged into a unified patient record. Since all extracted branches trace continuously from the ostium to a distal endpoint, they inherently share identical coordinates along common arterial trunks. When these tables are combined, the system merges overlapping branch data based on rounded (P_x, P_y, P_z) tuples. If multiple entries exist for the same spatial coordinate, the algorithm strictly retains the row with the highest %AS value. This conservative rule ensures that no maximum stenosis is hidden, preventing clinical underestimation of lesion severity where anatomical paths overlap.

Ultimately, Block 2 yields two structured outputs: the enriched individual branch datasets (containing g_d, A_{prox}, A_{dist}, A_{ref}, and %AS, among others) and the merged global coronary tree dataset. These comprehensive tables constitute the definitive quantitative input consumed by Block 3.

2.5. Block 3: Artery Labeling and CAD-RADS 2.0 Scoring

Block 3 translates the quantitative stenosis metrics produced in Block 2 into a standardized clinical classification using the Coronary Artery Disease Reporting and Data System (CAD-RADS 2.0) framework [13].

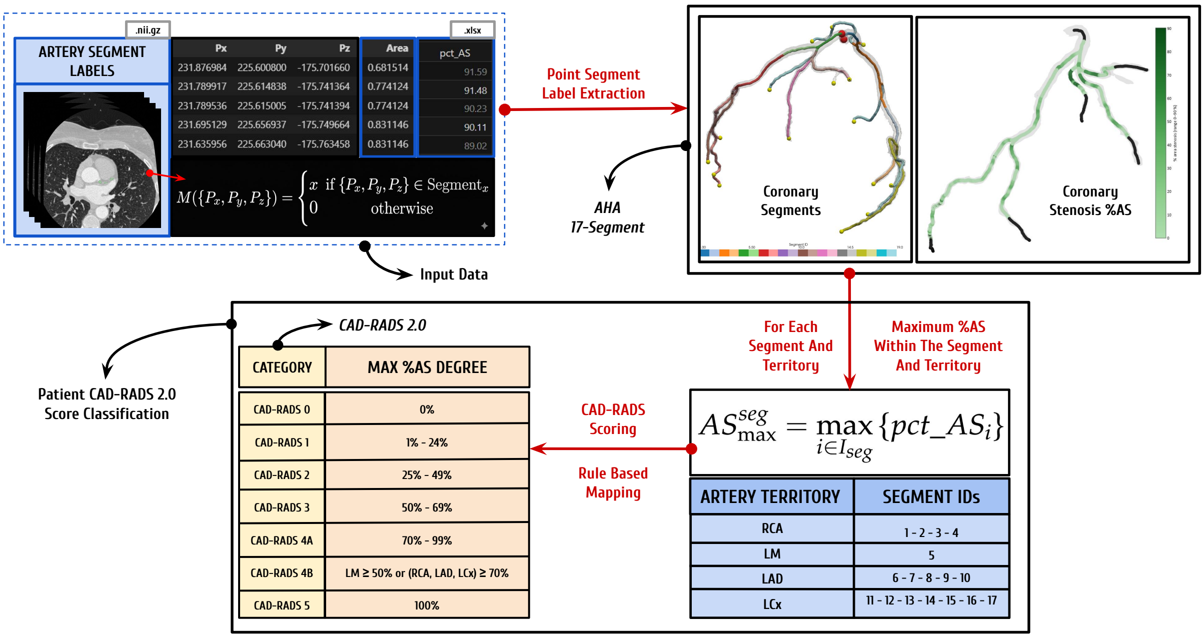

Figure 2.4: Block 3 workflow: segment aggregation based on the American Heart Association (AHA) 17-segment model and automated CAD-RADS 2.0 scoring.

As illustrated in Figure 2.4, this block ingests the merged patient-level stenosis table from Block 2 together with the anatomical segment labels inherited from Block 1. The pipeline first summarizes the stenosis severity at the individual segment level. These segments are subsequently grouped into the primary coronary territories (RCA, LM, LAD, and LCx), establishing the essential anatomical framework required for the CAD-RADS 2.0 evaluation. Finally, the system applies rule-based logic to derive the overall patient-level CAD-RADS 2.0 category and calculates the Segment Involvement Score (SIS), structuring these findings into the definitive outputs consumed by Block 4.

2.5.1. Anatomical Context: AHA 17-Segment Model

Reliable CAD-RADS 2.0 assignment requires mapping each evaluated centerline point to a recognized coronary territory. The pipeline adopts the 17-segment American Heart Association (AHA) model, a widely used reporting standard for regional coronary anatomy [4]. This choice is not only clinically established but also technically aligned with the voxel labeling scheme of the primary cohorts used in this thesis: integer labels in the ASOCA annotation volumes [16] are consumed directly as segment ID values without intermediate remapping. This direct compatibility preserves pipeline robustness and keeps segment-level aggregation consistent across patients.

Beyond individual anatomical segments, this model maps every branch into one of four primary coronary territories: the Right Coronary Artery (RCA), Left Main (LM), Left Anterior Descending (LAD), and Left Circumflex (LCX). This territorial grouping is a fundamental requirement of the CAD-RADS 2.0 scoring framework, which assigns the patient-level category from the most severe stenosis observed across these vessels [13]. Table 2.2 summarizes the complete segment-to-territory dictionary used throughout Block 3 and the clinical dashboard. Background voxels are encoded as segment 0 and are subsequently excluded from stenosis aggregation.

Table 2.2: AHA 17-segment dictionary used for anatomical labeling and CAD-RADS 2.0 territory grouping across RCA, LCA, left anterior descending (LAD), left circumflex (LCx), and left main (LM) territories. Adapted from Austen et al. [4].

ID	Abbreviation	Anatomical name	Territory
1	pRCA	Proximal RCA	RCA
2	mRCA	Mid RCA	RCA
3	dRCA	Distal RCA	RCA
4	rPDA	Right PDA	RCA
5	LM	Left Main	LM
6	pLAD	Proximal LAD	LAD
7	mLAD	Mid LAD	LAD
8	dLAD	Distal LAD	LAD
9	D1	First diagonal	LAD
10	D2	Second diagonal	LAD
11	pLCX	Proximal LCX	LCX
12	OM1	First obtuse marginal	LCX
13	dLCX	Distal LCX	LCX
14	OM2	Second obtuse marginal	LCX
15	lPDA	Left coronary PDA	LCX
16	L-ILB	LCX inferolateral branch	LCX
17	L-PLB	LCX posterolateral branch	LCX

2.5.2. Segment-Level Stenosis Aggregation

For each patient, all centerline rows in the Block 2 merged table that carry a valid `Segment_ID` are grouped by segment identifier using tabular aggregation. Within each segment, the algorithm retains the absolute maximum %AS, ensuring that focal lesions are not underestimated when multiple samples exist along the same anatomical region. Each segment is then assigned a discrete stenosis severity grade aligned with the CAD-RADS 2.0 luminal stenosis bins (see Table 2.3).

Any `Segment_ID` present in the input data but outside the 1–17 dictionary (for example, labels introduced by extended annotation schemes) is flagged as *unmapped*. These rows may still appear in exploratory outputs, but they are excluded from the standard territory groupings used for CAD-RADS 2.0 triage, preventing execution failure while making non-standard labels explicit.

2.5.3. Patient-Level CAD-RADS 2.0 Classification

Patient-level scoring follows the CAD-RADS 2.0 decision principles [13]. The algorithm first determines the highest segment-level stenosis grade from the aggregated table. This grade is derived from the maximum %AS observed across all evaluated segments and subsequently classified into the standard luminal categories detailed in Table 2.3. Total coronary occlusion (category 5) is assigned when any segment reaches 100% %AS.

Categories 4A and 4B distinguish high-risk patterns among severe stenoses. As established by the CAD-RADS 2.0 framework [13], category 4B is assigned when either (i) left-main (segment 5) stenosis is at least 50% %AS, or (ii) obstructive disease ($\geq 70\%$ %AS) is present in each of the three major territories (RCA, LAD, and LCX) defined in Table 2.4. If severe stenosis is present but neither of these criteria is met, the patient is classified as CAD-RADS 2.0 category 4A. Categories 0–3 are assigned directly from the highest segment grade when no category 4 or 5 rule overrides the result.

Table 2.3: CAD-RADS 2.0 luminal stenosis categories applied to maximum segment-level %AS. Adapted from Cury et al. [13].

Category	Max. %AS (segment)	Clinical interpretation (summary)
0	0%	No visible stenosis
1	1–24%	Minimal stenosis
2	25–49%	Mild stenosis
3	50–69%	Moderate stenosis
4A	70–99%	Severe stenosis without CAD-RADS 2.0 category 4B criteria
4B	70–99% (territory pattern) or LM $\geq 50\%$	High-risk severe disease pattern
5	100%	Total coronary occlusion

Table 2.4: CAD-RADS 2.0 triage territories per AHA 17-segment model. Adapted from Austen et al. and Cury et al. [4, 13].

Territory	Segment IDs	Role in automated scoring
RCA	1, 2, 3, 4	Obstructive/severe flags per vessel; contributes to three-vessel 4B rule
LM	5	Left-main $\geq 50\%$ %AS triggers CAD-RADS 2.0 category 4B
LAD	6, 7, 8, 9, 10	Obstructive/severe flags per vessel; contributes to three-vessel 4B rule
LCX	11, 12, 13, 14, 15, 16, 17	Obstructive/severe flags per vessel; contributes to three-vessel 4B rule

Finally, the system calculates the Segment Involvement Score (SIS), defined as the absolute number of assessable coronary segments exhibiting any measurable atherosclerotic plaque [13]. Block 3 concludes by exporting a comprehensive set of structured outputs: a segment-level stenosis summary, a patient-level CAD-RADS 2.0 report (detailing the assigned category, classification rationale, highest lesion location, territory flags, and SIS), and the geometrically enriched branch tables. All these generated metrics and records are specifically formatted to serve as the direct quantitative input for the interactive clinical visualization dashboard detailed in Block 4.

2.6. Block 4: Patient Visualization Dashboard

Block 4 constitutes the final user-facing layer of the automated workflow, functioning as an interactive clinical decision-support interface built with the Streamlit framework [32]. Whereas the preceding blocks operate as fully automated processing stages, this final block translates their geometric meshes and tabular records into an explorable visual environment. Specifically, the dashboard ingests the extracted centerline coordinates, cross-sectional areas, percentage area stenosis profiles, AHA segment labels, segment-level summaries, and the patient-level CAD-RADS 2.0 report.

Conceived as a rapid evaluation pop-up rather than a standalone diagnostic system, the interface synthesizes these artifacts into a single exploration session. The design attempts to avoid “black-box” behavior, recognizing that medical professionals highly value the transparency and explainability of automated tools. Consequently, every displayed score remains directly traceable to the underlying geometric measurements.

Within the clinical workflow of Hospital de la Santa Creu i Sant Pau, this dashboard is designed to integrate seamlessly with the recently developed patient prioritization system [14]. By accessing this interface for prioritized cases, cardiologists can obtain a rapid, detailed overview to efficiently rule out low-risk patients or decide if a deeper inspection of the raw coronary computed tomography angiography (CCTA) scans is required.

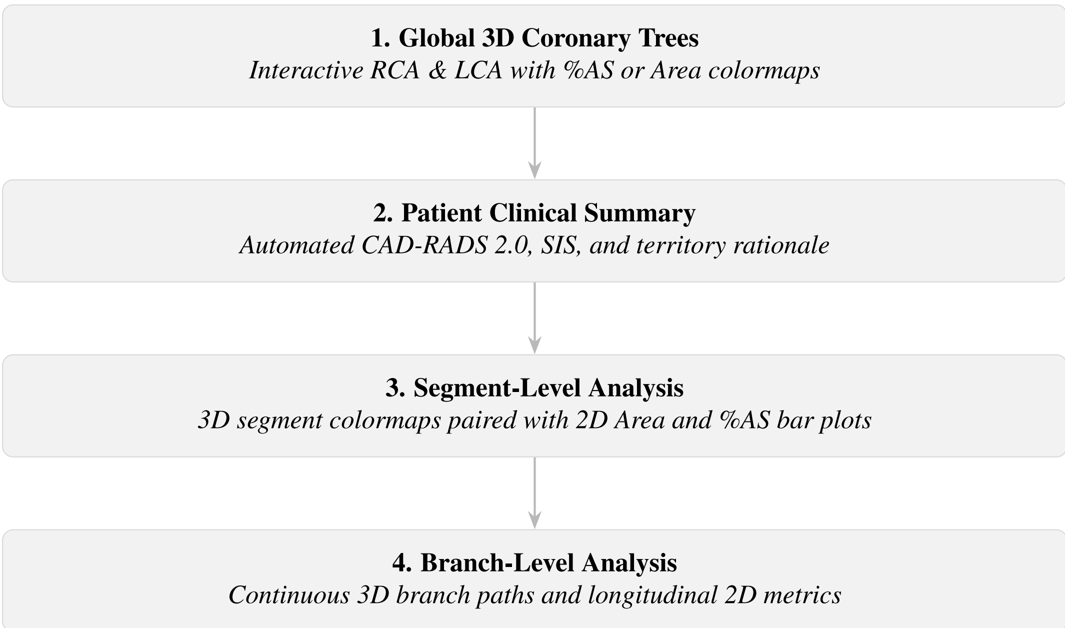

Figure 2.5: Block 4 user interface layout and interaction flow: from global RCA and LCA anatomy with %AS colormaps, through CAD-RADS 2.0 and segment involvement score (SIS) summary, to segment-level and branch-level metric validation.

As illustrated in Figure 2.5, the Streamlit dashboard is organized into four vertically stacked modules that guide the clinical review process from global anatomy to vessel-level metrics. Module 1 opens the session with interactive three-dimensional renderings of the global RCA and LCA, which can be colored by either %AS or cross-sectional area. Module 2 immediately follows with the patient clinical summary, reporting the automated CAD-RADS 2.0 category, the Segment Involvement Score (SIS), and the territory-level rationale [13]. Below these high-level overviews, Modules 3 and 4 offer dedicated segment-level and branch-level exploration. In both layers, localized three-dimensional anatomical views are shown alongside their corresponding two-dimensional metric profiles, allowing a comprehensive evaluation of the affected vessels.

Module 3: Segment Level Analysis

The third module focuses on the detailed analysis of individual segments, following the standard AHA 17-segment framework. The three-dimensional view initially displays the entire artery colored by segment identity. When a user selects a specific segment from either the RCA or LCA, the three-dimensional map highlights this selection so the user clearly sees the region under analysis. Concurrently, two synchronized two-dimensional bar plots update to display the cross-sectional area and %AS profiles. This allows the user to visualize exactly how the area and stenosis behave along the length of the chosen segment. To ensure the severity calculations are transparent and easy to verify, the charts explicitly mark the point of maximum stenosis alongside the reference areas used in Equation (2.2). This combination of three-dimensional models and two-dimensional charts is supported by visualization research [6]. The three-dimensional anatomy provides clear spatial context, while the two-dimensional profiles eliminate visual overlap, making it significantly easier to locate lesions and accurately read the metrics.

Module 4: Branch-Level Analysis

The fourth module applies the same dual-visualization principle as Module 3, but expands the scope to the continuous ostium-to-endpoint paths extracted in Block 1. Upon selecting a specific branch, the three-dimensional view highlights its full anatomical trajectory and flags the peak stenosis location. Below this, longitudinal area and %AS bar plots chart the entire branch length in a strict proximal-to-distal order. Because the stenosis quantification in Block 2 is computed independently along these paths, this view exposes the exact continuous geometric context from which the reference windows and %AS values were derived.

In Modules 3 and 4, explicitly plotting the peak stenosis alongside its bilateral reference samples guarantees full algorithmic transparency. This design allows cardiologists to rapidly validate the automated findings against their clinical intuition before deeper inspection of the raw CCTA data. Full-page screenshots demonstrating these dashboard interfaces are provided in Section A.5.

2.7. Experimental Design and Validation Framework

Validating an automated clinical decision-support pipeline requires complementary evidence at multiple levels of abstraction. As established in the project objectives (Section 1.5), the primary goal of this thesis is to construct and integrate the complete computational workflow, rather than to maximize or optimize the clinical accuracy of stenosis quantification.

Because of anatomical limitations discussed in Section 4.2, the pipeline overestimates stenosis severity in complex regions. Consequently, the current iteration of the pipeline does not possess reliable clinical predictive value. To align with the project scope, clinical stenosis accuracy and CAD-RADS 2.0 concordance against real-world patient diagnoses are explicitly excluded from the evaluation design.

Instead, this thesis adopts a three-tier validation framework: **(i)** operational validation on real patient cohorts (ASOCA and MACS-18) to assess algorithmic robustness and geometric extraction precision; **(ii)** controlled algorithmic verification on synthetic phantoms with analytical ground truth to validate the underlying mathematical logic; and **(iii)** a pilot clinical usability study using the System Usability Scale (SUS), conducted with medical experts at Hospital de la Santa Creu i Sant Pau, to evaluate the interactive Block 4 interface. Numerical outcomes are reported in Chapter 3; interpretation of limitations and future directions is reserved for Chapter 4.

2.7.1. Real-Data Evaluation Metrics and Acceptance Criteria

The first validation tier applies a shared 40-patient cohort (20 normal and 20 diseased anatomy) defined in Section 2.2. The real-data metrics in Table 2.5 were defined to evaluate pipeline reliability, geometric extraction quality, and runtime scalability on ASOCA and MACS-18. They measure whether the workflow runs successfully, whether Block 1 recovers plausible coronary topology, and whether automated processing remains practical at cohort scale.

Table 2.5: Real-data evaluation metrics used in Sections 2.7.2 and 2.7.3.

Metric	Definition	Measurement
ESR	Execution Success Rate: fraction of cases completing Blocks 1–4 without fatal error	Automated pipeline logs
MPE	Mean Endpoints per Patient: average skeleton endpoint count per case	Block 1 output (descriptive topology richness)
EISR	Endpoint Identification Success Rate: correct endpoints / all detected endpoints	Manual visual audit per endpoint (ASOCA)
OISR	Ostium Identification Success Rate: correctly placed RCA and LCA ostium points out of 2 expected per patient	Manual visual inspection vs. aortic origin
CESR	Centerline Extraction Success Rate: successful ostium-to-endpoint paths / attempted extractions	Automated computation from Block 1 extraction records
Runtime	Wall-clock seconds per block and total per patient	Logged timestamps

Table 2.6 assigns each real-data metric a *good*, *neutral*, or *poor* band to simplify interpretation of the Chapter 3 results: *good* meets the expected level for this thesis, *neutral* denotes an acceptable but imperfect outcome, and *poor* marks a clear limitation. The thresholds were proposed by the author in consultation with clinical supervisors at Hospital de la Santa Creu i Sant Pau and should be treated as practical project benchmarks rather than official medical guidelines.

Table 2.6: Acceptance bands for real-data pipeline metrics (ESR, OISR, CESR, and EISR as defined in Table 2.5).

Metric	Good	Neutral	Poor
ESR	$\geq$95%	85–94%	$<$85%
OISR / CESR	$\geq$85%	75–84%	$<$75%
EISR (ASOCA only)	$\geq$85%	75–84%	$<$75%
Mean Total Runtime	$<$500 s	500–900 s	$>$900 s

2.7.2. Workflow Validation on Real Datasets: ASOCA

The ASOCA resource [16] served as the development reference for Blocks 1–4. After pipeline stabilization, the full 40-patient ASOCA cohort was processed end-to-end with identical configuration to that described in Sections 2.3 to 2.6.

Goal: Evaluate geometric precision, runtime efficiency, and execution success on the initial public benchmark masks.

Procedure: For each patient, the pipeline was executed without manual intervention. The metrics defined in Table 2.5 (ESR, EISR, OISR, CESR, and per-block runtimes) were recorded for normal, diseased, and combined subgroups. Both ostium and endpoint correctness were assessed via manual visual inspection: ostium accuracy was judged by overlaying detected ostium coordinates on the 3D coronary surface and comparing them to the anatomical origins at the aortic root, while each detected endpoint was individually verified for topological validity. Finally, CESR was computed automatically from Block 1 extraction records. Comparative numerical results for this experiment are reported in Section 3.1.

2.7.3. Workflow Validation on Real Datasets: MACS-18

The identical pipeline was re-executed on the expert-refined MACS-18 lumen masks for the same 40 patient identifiers [1]. Only the input segmentation differed; segment label volumes and pipeline hyperparameters were held constant.

Goal: Determine whether the improved lumen connectivity in MACS-18 affects geometric extraction robustness, topological endpoint density, and computational cost relative to ASOCA.

Procedure: To enable a direct comparison, the same 40 patients were each processed through the pipeline without manual intervention. The applicable metrics defined in Table 2.5 (ESR, MPE, OISR, CESR, and per-block runtimes) were recorded to directly contrast both cohorts. The evaluation protocol mirrored that of Section 2.7.2: ostium correctness was assessed via manual visual inspection against the anatomical origins at the aortic root, and CESR was computed automatically from Block 1 extraction records. Comparative numerical results for this paired experiment are reported in Section 3.1.

2.7.4. Geometry and Quantification Verification: Synthetic Data

Clinically, stenosis quantification would typically be evaluated by comparing algorithmic CAD-RADS 2.0 categories against expert radiological reports or by assessing the maximum stenosis within specific arterial segments. However, as previously noted, natural anatomical irregularities cause the pipeline's current area-based formula [8, 17] to overestimate stenosis severity, limiting its direct clinical predictive value on real patient cohorts.

Following consultation with clinical supervisors at Hospital de la Santa Creu i Sant Pau, verification was instead conducted in a controlled setting using two programmatically generated vessel models (healthy and diseased). These idealized shapes isolate the formula from anatomical confounders and were processed through the full pipeline with the same 10 mm reference half-window applied to the clinical cohorts.

Goal: Verify the underlying mathematical logic and geometric accuracy of the quantification algorithms by comparing the pipeline outputs against an absolute analytical ground truth, completely isolated from real-world anatomical noise. Numerical outcomes are reported in Section 3.2.

Synthetic Model Design: Both masks occupy a $100 \times 100 \times 100$ voxel grid with a uniform 1.0 mm spacing in all three dimensions. Each model is a capped cylinder along the Z-axis (from $Z = 15$ to $Z = 85$) with hemispherical end caps to ensure closed surfaces compatible with VMTK cross-sectioning.

- **Synthetic 1 (Healthy):** Constant radius $R = 10\,\text{mm}$. Theoretical cross-sectional area $A = \pi R^2 \approx 314.16\,\text{mm}^2$ and maximum %AS $= 0\%$.
- **Synthetic 2 (Diseased):** Same base radius with a symmetric cosine narrowing between $Z = 40$ and $Z = 60$, reaching a minimum radius $R_{\min} = 5\,\text{mm}$ at $Z = 50$. Theoretical maximum area is $\approx 314.16\,\text{mm}^2$, minimum area is $\approx 78.54\,\text{mm}^2$, and maximum %AS $= 75\%$.

The stenosis radius profile follows the equation:

$$R(z) = R_{\min} + (R_{\text{base}} - R_{\min}) \frac{1 - \cos\left(\pi \frac{z - Z_{\text{mid}}}{H}\right)}{2}, \quad z \in [40, 60], \tag{2.4}$$

with base radius $R_{\text{base}} = 10\,\text{mm}$, minimum radius $R_{\min} = 5\,\text{mm}$, midpoint $Z_{\text{mid}} = 50$, and half-width $H = 10\,\text{mm}$.

Procedure: For each model, the pipeline's peak %AS and cross-sectional area samples were directly compared against the analytical ground truth. Errors are reported as absolute differences in percentage points (for %AS) and mm^2 (for area). Longitudinal area and %AS profiles were also extracted for visual verification. Table 2.7 assigns each metric a *good*, *neutral*, or *poor* band using the same interpretive logic as Table 2.6: %AS cutoffs follow the 5–10% inter-observer variability range [17], and area cutoffs in mm^2 reflect mesh-discretization tolerance on ideal models. These values were agreed with clinical supervisors and should be treated as practical project benchmarks.

Table 2.7: Acceptance thresholds for synthetic model verification.

Metric	Good	Neutral	Poor
Max %AS Absolute Error	<5%	5–10%	>10%
Mean Area Error (Healthy)	$<6\,\text{mm}^2$	$6\text{–}16\,\text{mm}^2$	$>16\,\text{mm}^2$
Min-Area Error (Stenosis)	$<5\,\text{mm}^2$	$5\text{–}15\,\text{mm}^2$	$>15\,\text{mm}^2$

2.7.5. User Experience and Interface Usability Evaluation

The third validation tier evaluates the interactive dashboard developed in Block 4 from the end user's perspective. Medical experts at Hospital de la Santa Creu i Sant Pau took part in a pilot System Usability Scale (SUS) study [7] of this interface. The study focused on usability, clarity, and overall clinical acceptance.

Procedure: Three medical experts attended individual in-person or online sessions comprising a brief project introduction, hands-on interaction with the full Block 4 dashboard on representative outputs, and the standard 10-item SUS Likert questionnaire (1–5; full wording in Section A.2) with optional qualitative feedback. Responses were scored 0–100 using the standard SUS formula [7]. The cohort mean and standard deviation are reported in Section 3.4 and interpreted using the score bands in Table 2.8.

Table 2.8: SUS score interpretation bands. Adapted from Bangor et al. and Brooke [5, 7].

SUS Score	Grade	Interpretation
≥80	Excellent	High usability
68–79	Good	Above-average; acceptable for release consideration
50–67	OK	Improvement needed
<50	Poor	Not acceptable

Chapter 3

RESULTS

This chapter reports the three-tier validation outcomes (Section 2.7), classified using Tables 2.6–2.8 and discussed in Chapter 4.

3.1. Pipeline Execution: ASOCA vs. MACS-18

Figure 3.1 introduces the paired ASOCA versus MACS-18 comparison using patient Normal_8, a representative normal case from the shared 40-patient cohort. The figure contrasts Block 1 centerline extraction, endpoint detection, ostium identification, and surface mesh generation when only the input lumen mask differs between datasets.

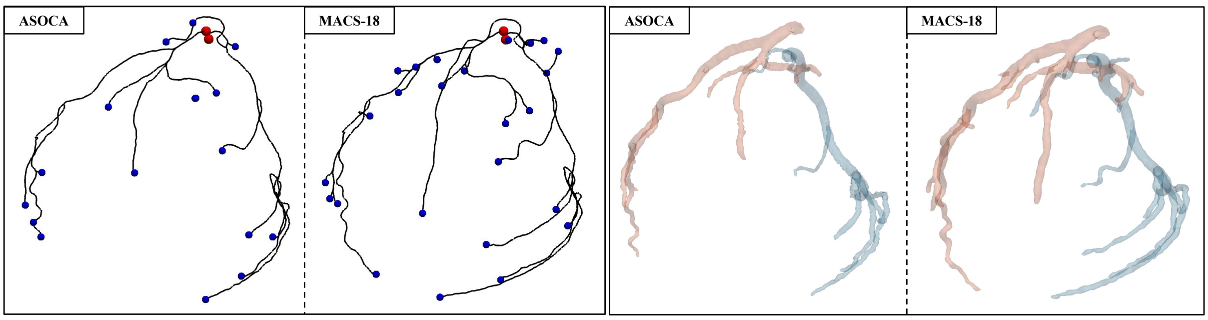

Figure 3.1: Block 1 outputs for Normal_8 in four side-by-side panels. Left pair (ASOCA [Automated Segmentation of Coronary Arteries], then MACS-18 [Multiclass Anatomical Coronary Segmentation]): centerline paths with red ostium points and blue branch endpoints. Right pair (ASOCA, then MACS-18): smoothed lumen surfaces with light-red LCA and light-blue RCA.

As stated in Section 2.2.1, MACS-18 provides more anatomically detailed lumen masks than ASOCA. For the same patient (e.g., Normal_8), this difference can be observed in both the centerline and surface views, where MACS-18 yields sharper branch definition.

On ASOCA masks, every sample completed Blocks 1–4 without fatal error (ESR = 100%, *good*). The cohort averaged 17.1 detected endpoints per patient, with 89.77% correctly identified (EISR, *good*). Ostium placement reached 85.0% success (OISR, *good*), and centerline extraction succeeded in 83.19% of attempted paths (CESR, *neutral*). Mean total runtime was 346.9 s (*good*).

On MACS-18 masks, every sample completed Blocks 1–4 without fatal error (ESR = 100%, *good*). Mean endpoints per patient increased to 27.7, consistent with the richer distal branching in the refined masks. Ostium success was 73.75% (OISR, *poor*), and centerline extraction improved to 95.62% (CESR, *good*). Mean total runtime was 576.2 s (*neutral*).

The next tables summarize endpoint counts (Table 3.1), ostium and centerline accuracy (Table 3.2), and mean Block 1–4 runtimes (Table 3.3) for the shared 40-patient ASOCA and MACS-18 cohorts. Extended distribution charts appear in Section A.3.

Table 3.1: Endpoint summary statistics by cohort and subgroup.

Cohort	Subgroup	n	Mean	Std	Min	Max
ASOCA	All	40	17.1	4.78	10	32
ASOCA	Normal	20	18.2	5.61	12	32
ASOCA	Diseased	20	16.0	3.58	10	22
MACS-18	All	40	27.7	10.0	12	64
MACS-18	Normal	20	30.8	11.6	17	64
MACS-18	Diseased	20	24.7	7.15	12	42

Table 3.2: OISR and CESR by ASOCA and MACS-18 cohort and subgroup (%).

Cohort	Subgroup	Ostium (%)	Centerline (%)
ASOCA	All	85.0	83.19
ASOCA	Normal	85.0	83.18
ASOCA	Diseased	85.0	83.21
MACS-18	All	73.75	95.62
MACS-18	Normal	75.0	96.87
MACS-18	Diseased	72.5	94.04

Table 3.3: Mean pipeline runtimes for ASOCA and MACS-18 by block (seconds).

Cohort	Block 1	Block 2	Block 3	Block 4	Total
ASOCA	145.8	164.6	35.2	0.16	346.9
MACS-18	159.8	357.7	57.9	0.02	576.2

3.2. Geometric and Algorithmic Accuracy via Synthetic Data

Block 2 quantification was verified on the two synthetic vessel models defined in Section 2.7.4. The analytical ground truth for lumen area and %AS specified in that section isolates algorithmic error from the anatomical variability of the real cohorts in Section 3.1. Both models were processed through the full pipeline; Figure 3.2 shows the resulting tube reconstructions with extracted centerlines.

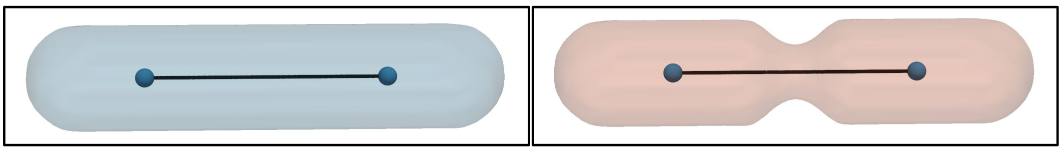

Figure 3.2: Extracted centerlines on the synthetic validation models. Left: `synthetic_1` (healthy, uniform radius). Right: `synthetic_2` (diseased, mid-vessel stenosis).

As Table 3.4 below shows, pipeline execution on both synthetic models yields absolute errors that all fall within the *good* acceptance bands of Table 2.7. For each metric, the analytical ground truth defined in Section 2.7.4 is compared against the computed Block 2 output; the largest deviations occur on `synthetic_2`, with a maximum %AS error of 1.31% and a maximum area error of 4.50 mm^2. Additional visualizations and information about these results, including longitudinal area and %AS profiles, are provided in Section A.4.

Table 3.4: Synthetic validation outcomes ($W = 10$ mm reference half-window). %AS and cross-sectional area errors versus analytical ground truth.

Model	Metric	Theoretical	Computed	Error	Unit
Healthy	Max %AS	0.0	0.01	0.01	%
Healthy	Mean area	314.16	316.50	2.34	mm^2
Diseased	Max %AS	75.0	73.69	1.31	%
Diseased	Max area	314.16	318.66	4.50	mm^2
Diseased	Min area	78.54	82.55	4.01	mm^2

3.3. Coronary Artery Dashboard Visualization Output

Running validated pipeline outputs through Block 4 produces the clinical visualization views described in Section 2.6. For representative patient cases, these include global RCA and LCA renderings, the automated CAD-RADS 2.0 report with SIS, and synchronized segment- and branch-level area and %AS profiles. Full module-by-module screenshots are collected in Section A.5.

3.4. User Experience and Interface Usability Evaluation

Following the medical expert evaluation in Section 2.7.5, Tables 3.5 and 3.6 report the raw Likert responses and computed SUS scores (questionnaire wording in Section A.2).

Table 3.5: SUS raw Likert responses (1–5). P1–P3 denote participants 1–3.

Item	P1	P2	P3
Question 1	4	5	4
Question 2	1	3	1
Question 3	3	5	4
Question 4	3	3	1
Question 5	3	4	4
Question 6	3	2	2
Question 7	2	4	4
Question 8	3	2	1
Question 9	3	3	4
Question 10	3	2	1

Table 3.6: SUS score summary by participant and metrics.

Participant	Score
Participant 1	55.0
Participant 2	72.5
Participant 3	85.0
Mean	70.83
Std. dev.	15.07

The mean SUS score was 70.83 (SD = 15.07), placing the dashboard in the *Good* band ($\geq$68) of Table 2.8.

Chapter 4

DISCUSSION

This chapter interprets the validation results from Chapter 3 to evaluate the pipeline's operational reliability, geometric accuracy, and clinical usability. It then discusses the primary technical and anatomical limitations of the current automated stenosis quantification, concluding with a prioritized future work agenda driven by expert feedback from clinical experts at Hospital de la Santa Creu i Sant Pau.

4.1. Analysis of the Results

The Topological Trade-Off in MACS-18

When processing the 40 patients, the pipeline remained stable for both the ASOCA and MACS-18 cohorts, achieving an Execution Success Rate of 100%. The richer anatomical fidelity of MACS-18 clearly improves the topology recovery in Block 1. As shown in Table 3.1, the mean detected endpoints increased from 17.1 to 27.7 per patient, and the Centerline Extraction Success Rate improved from 83.19% to 95.62% (Table 3.2). These gains suggest that MACS-18 better preserves distal continuity and lumen details, allowing the pathfinding algorithm to traverse segments that were previously disconnected or lost (Figure 3.1).

However, observing the results, it seems that this same anatomical richness negatively affects ostium identification. The Ostium Identification Success Rate fell from 85.0% on ASOCA to 73.75% on MACS-18 (Table 3.2). This decrease could be explained by the higher complexity of the vessel skeleton. Because MACS-18 captures more distal branches, the skeleton contains more junctions and complex structures near the root. Consequently, the automated detection method might misclassify a proximal branch junction as the true ostium candidate.

Overall, while the MACS-18 masks provide a much more precise geometry that improves centerline extraction, our current ostium selection technique struggles with this increased structural density. This suggests that, rather than the detailed masks being a problem, our ostium detection method should be optimized to handle them properly. By improving this specific step, a potential upgrade that will be discussed further in Section 4.3, the pipeline could fully leverage the high geometric precision of datasets like MACS-18.

Computational Efficiency and Runtime Discrepancies

As reported in Table 3.3, the mean total pipeline runtime increased from 346.9 seconds on ASOCA to 576.2 seconds on MACS-18. The main cause of this difference is Block 2, responsible for the geometric quantification, which rose from 164.6 seconds to 357.7 seconds. This increase is a logical result of the system design. At each centerline point, the pipeline calculates a cross-sectional area by slicing the 3D surface mesh. Therefore, the total time depends directly on the amount of data processed. Because

MACS-18 captures more branches and longer distal paths (Table 3.1), the algorithm simply has more valid geometry to analyze, multiplying the total area calculations required. The remaining blocks only showed minor time increases due to handling slightly larger files.

Consequently, the reported runtimes are compatible with a background processing workflow rather than with on-demand computation during a live session. In the intended hospital use case, the pipeline can be executed after segmentation is available and before the clinical expert opens the dashboard, storing the quantification outputs for each prioritized patient. When the doctor later requests the pop-up review, the interface can load precomputed results directly, without repeating the full Blocks 1–3 chain. Even under this model, the higher computational cost of MACS-18 remains relevant: as segmentation quality improves, quantification efficiency should be optimized in parallel to keep background processing practical at scale.

Algorithmic Verification via Synthetic Models

The primary objective of the synthetic experiments (Section 3.2) is to verify the mathematical correctness of our quantification formulas without the interference of anatomical noise or complex bifurcations that could bias the results. As shown in Table 3.4, this verification provides clear evidence that the Block 2 logic is consistent. All calculated errors for both synthetic models fall well within the *good* acceptance bands defined previously in Table 2.7. Even on the diseased model, the maximum percentage area stenosis error is only 1.31%, and the maximum area error is just 4.50 mm^2.

Furthermore, the extended longitudinal profiles provided in Section A.4 clearly visualize the artificial narrowing generated in the vessel. In this controlled scenario, the selected window size successfully captures healthy reference points, ensuring a realistic and accurate comparison. Therefore, the tiny residual errors observed should not be seen as a flaw in the stenosis equations themselves. Instead, this small deviation is caused by the internal VMTK functions [19] used to extract the cross-sectional area at each point. The process of converting analytically defined shapes into voxelized surfaces, smoothing the mesh, and calculating discrete slices along the centerline introduces minor numerical noise. This slightly impacts the area measurement and naturally carries over to the final stenosis percentage calculation.

Ultimately, these synthetic tests verify the fidelity of the implementation. They confirm that the software can reproduce known geometries with almost zero deviation. By proving that the core formula works perfectly in an ideal environment, it becomes clear that any inaccuracies encountered in real patient data are not due to mathematical errors. Instead, they arise from the challenge of applying a fixed reference window to complex human anatomy and the inherent difficulty of automatically locating true healthy reference points in a diseased vessel. These specific structural limitations will be analyzed in detail in Section 4.2.

Clinical Usability and Acceptance

The pilot SUS evaluation detailed in Section 3.4 yielded a mean score of 70.83, placing the dashboard in the *Good* acceptability band of Table 2.8. Because this interface represents a first iteration, it was expected that the tool would not be flawless. The primary goal was to establish a functional baseline that effectively communicated the core concept of the pipeline, with the intention of refining it over time. This experimental nature helps explain the diverse opinions and the spread in acceptance observed among the three clinical experts, whose individual scores ranged from 55.0 to 85.0 (Table 3.6). A closer look at the item-level responses (Table 3.5) provides deeper insight into this variation.

Rather than merely summarizing the numerical results, it is important to interpret what they mean for the dashboard's design. The most positive signals relate to the system's perceived clinical utility and visual integration. Strong ratings on the intent for frequent use (Question 1) and the integration of 3D meshes with 2D plots (Question 5) indicate that the core concept resonates well with clinical needs. Conversely, responses regarding diagnostic confidence (Question 9) and the need for technical support

(Question 4) were more mixed. Referencing the full questionnaire provided in Section A.2, these varied answers suggest that while the visual layout is highly appreciated, the tool requires further refinement before clinicians can feel completely autonomous and confident using it without prior training; specific interface improvements proposed by participants are outlined in Section 4.3.3.

Beyond the scores, the session sparked valuable debate among the cardiologists and surfaced concrete ideas to improve future visualization iterations, developed in Section 4.3. Physicians' overall impressions were highly positive, affirming substantial practical utility within the clinical ecosystem of Hospital de la Santa Creu i Sant Pau.

4.2. Technical and Anatomical Limitations

Anatomical Confounders, Tapering, and Area Overestimation

The core limitation of this thesis is not operational failure, but geometric sensitivity when processing complex real coronary anatomy. In Block 2, stenosis is estimated by comparing each point's local cross-sectional area with a bilateral healthy reference derived from adjacent points along the branch vessel. While this approach works perfectly on ideal synthetic tubes (Section 3.2), it struggles with natural anatomical variations: coronary arteries taper from proximal to distal, and bifurcations create zones where the vessel naturally widens. Because the reference window is recomputed continuously at every centerline point, overestimation arises when a point's proximal or distal window falls inside one of these wider morphological regions or along a highly tapered segment.

Geometrically, the cross-sectional area is calculated using a plane perpendicular to the centerline's tangent. At bifurcations, where the centerline curves to redirect into a new branch, this perpendicular slice can capture an artificially elongated section. If the algorithm selects one of these controversially large, non-representative points as the "healthy" baseline, the reference area becomes heavily inflated. Consequently, when the algorithm evaluates the points immediately downstream, it falsely interprets them as pathologically narrow, leading to a drastic overestimation of the percentage area stenosis.

Figure 4.1 illustrates this phenomenon. As observed in the bar plots, the percentage area stenosis (orange bars) spikes significantly for the points located just after the bifurcation.

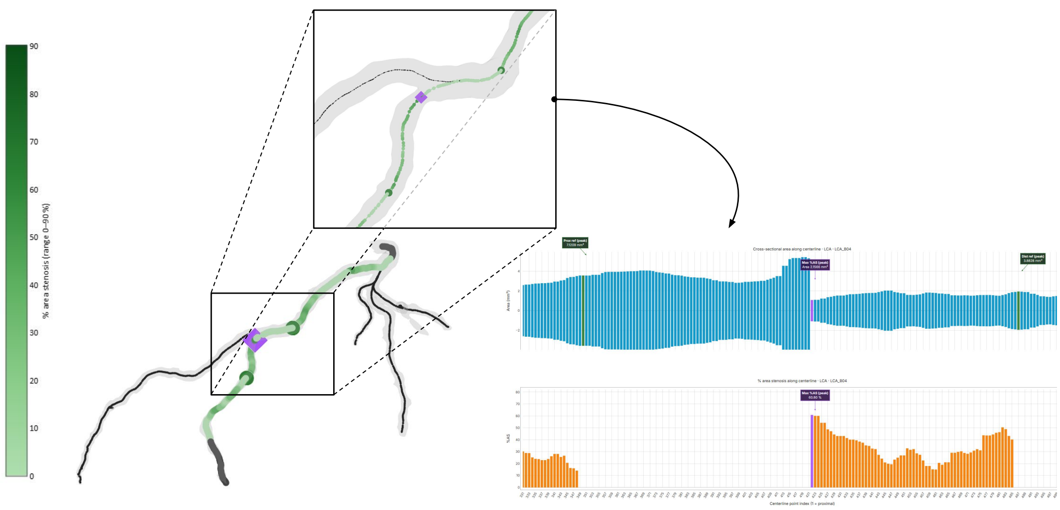

Figure 4.1: Area overestimation near a coronary bifurcation. The 3D model maps %AS (green colormap); the inset marks the maximum stenosis point (purple diamond) and automatic proximal/distal references (green circles). Bar plots show cross-sectional area (blue) and %AS (orange), including the artificial spike after the bifurcation.

However, after a certain distance, this overestimation disappears and the values abruptly drop back to a normal baseline. This stabilization occurs because the proximal reference window has finally moved completely out of the large main vessel and into the smaller sub-vessel, finding a truly representative healthy reference. A similar measurement distortion can happen when the vessel remodels in a non-circular shape due to plaque. Since the cross-sectional area reduces complex, irregular, or eccentric shapes into a single numerical value, it can sometimes fail to reflect the true anatomical reality of the lumen.

Because the final CAD-RADS 2.0 scoring relies strictly on the maximum stenosis values found across the vessels, this localized overestimation directly infects the final summary block. Consequently, the pipeline produces non-representative CAD-RADS 2.0 categories that are driven by normal vessel geometry rather than true pathological severity. For this reason, in terms of CAD-RADS 2.0 classification, the current version of the tool lacks autonomous predictive value and cannot independently represent the patient's actual condition. However, this is a highly fixable issue. By optimizing the quantification block and exploring alternative mathematical methods to identify truly healthy reference values, this problem can be resolved. In Section 4.3, we will present a specific, mathematically robust proposal discussed with clinical professionals at Hospital de la Santa Creu i Sant Pau to address this exact limitation in future iterations.

Rigidity of the Baseline Reference Window

Closely tied to the anatomical confounders discussed previously is the algorithmic rigidity used to locate these reference areas. In our current quantification method (Block 2), the reference window size (W) is one of the most critical parameters. Throughout this thesis, we utilized a fixed reference half-window of $W = 10\,\mathrm{mm}$, as defined and justified in Section 2.4. This value is an empirically motivated constant, chosen as a pragmatic compromise to enable reproducible batch processing and closed-form reference estimation. However, its rigidity is precisely what triggers the overestimation problems at bifurcations and tapered vessels. Because the algorithm strictly looks 10 mm upstream and downstream, it is entirely blind to the local context. It cannot distinguish whether those specific distances land on healthy tissue, diffuse disease, calcification, or a morphological branch expansion.

In routine clinical practice at Hospital de la Santa Creu i Sant Pau, cardiologists avoid this issue through visual judgment. They manually select reference sites within the same vessel segment, deliberately avoiding bifurcation areas, ostial zones, and lesion shoulders to find a truly healthy baseline. Replicating this expert human judgment automatically is substantially harder. A fixed 10 mm window cannot adapt to varying lesion lengths or branch calibers. For example, short side branches might not have enough length to satisfy the 10 mm search, while extensively diseased segments might force the window to sample pathological tissue instead of a healthy baseline. In both scenarios, the reference area becomes biased, leading to inaccurate stenosis percentages.

Because the algorithm currently lacks the context-awareness to dynamically adjust this window, the stenosis outputs on real cohorts should be interpreted as exploratory quantitative maps rather than definitive diagnostic metrics. Fortunately, after evaluating these results with clinical experts at Hospital de la Santa Creu i Sant Pau, we established a clear path forward. We have outlined a specific proposal to replace this rigid parameter with adaptive methods for finding true healthy reference values, which will be discussed in detail in Section 4.3.

Challenges in Automated Ostium Identification

Moving beyond the geometric limitations of area quantification, another structural challenge lies at the very beginning of the vessel: the ostium. As detailed in Section 2.3, our current automated heuristic identifies the ostium to define the complete vascular paths from the aortic root down to each endpoint. Overall, the metrics obtained for this step are actually quite positive, achieving an Ostium Identification

Success Rate of 85.0% on ASOCA and 73.75% on MACS-18 (Table 3.2). However, as these results indicate, the current technique does not always detect the exact anatomical root perfectly, occasionally struggling when faced with highly complex vessel structures.

It is important to clarify that an imprecise ostium placement does not break the execution logic of the pipeline. The geometric quantification in Block 2 can still successfully calculate cross-sectional areas and local stenosis percentages along whatever centerline path it receives. Nevertheless, because the system organizes the downstream branches starting from this anchor point, a misplaced ostium does influence the final reporting. An incorrect starting point might cause the omission of a proximal segment or slightly alter the way branches are grouped.

During the development of this thesis, the priority was to build a fully functional, end-to-end pipeline rather than exclusively optimizing this single detection step. Therefore, while the current success rates are solid, they do not represent the ceiling of what the algorithm can achieve. To ensure that the topological anchoring matches the high quality of the downstream quantification, we will propose strong alternative solutions and improved rules for identifying the ostium in Section 4.3.

Usability Study Sample Size

Finally, the clinical evaluation must be interpreted within its methodological limits. With only three clinical experts from a single institution participating (Sections 2.7.5 and 3.4), the System Usability Scale evaluation is strictly an exploratory pilot. Consequently, the resulting scores cannot support broad statistical inference.

Nevertheless, this sample size was perfectly appropriate for the scoped objectives of this bachelor thesis. Prioritizing deep qualitative feedback from specialized cardiologists provided immense value. The resulting clinical consensus on key areas like transparency, manual reference control, and validation grounded in medical imaging proved much more informative for future designs than expanding the cohort simply to achieve statistical significance.

4.3. Future Work

Building upon the limitations identified in the previous section, the following directions outline the next steps for refining the pipeline. These proposed improvements stem from two main sources: the need to directly address the specific geometric and technical constraints discovered during development, and the valuable practical feedback gathered from cardiologists at Hospital de la Santa Creu i Sant Pau during the evaluation sessions.

4.3.1. Algorithmic and Pipeline Enhancements

As previously discussed, the current version of the pipeline represents a first iteration designed to successfully integrate multiple processing blocks into a single functional tool. Moving forward, future work must focus on the individual optimization of each specific block. Refining these steps independently will yield much higher precision across the entire system and provide the necessary mechanisms to overcome the geometric and technical limitations mentioned in the previous sections.

Centerline Resolution Optimization

Clinical experts noted that extremely dense sampling is often unnecessary for routine triage. The current extraction granularity of 0.1 mm yields rich plots but significantly increases the computational cost of the algorithm. An adaptive resampling policy with a 2 mm default spacing, preserving fine sampling exclusively around suspected lesions, could drastically reduce processing times without sacrificing clinical readability.

Plaque Versus Stenosis Quantification

Clinical experts emphasized a crucial medical distinction that the present pipeline does not yet model. From a clinical perspective, doctors first look for the presence of plaque and then evaluate if that plaque is actively causing a stenosis. Plaque is simply the physical accumulation of calcium or cholesterol, whereas stenosis is the actual narrowing of the vessel. Importantly, a patient can have plaque accumulations that do not result in any significant narrowing. Because specialists inherently search for detected plaque zones to assess if they cause a relevant obstruction, integrating automated plaque detection and visualization would be exceptionally useful. Adding this feature alongside the current percentage area metrics would better mirror natural clinical reasoning.

Ostium Identification And Adaptive Reference Selection

To address the root anchoring challenges, the current heuristic rules for ostium detection should be upgraded. One highly promising alternative is implementing Artificial Intelligence models, since the aorta is a massive and visually distinct structure that a trained network can easily recognize. Another option is developing advanced mechanisms to directly analyze the initial segmentation and identify exactly which endpoint connects to the aortic root. Alongside these ostium improvements, the algorithm must update how it selects healthy reference areas. The system currently relies on a fixed distance window. A much better alternative would be to identify the healthy reference purely within the boundaries of the same vessel segment, exactly mimicking how human specialists operate. Furthermore, future iterations of the interface should allow the clinical expert to manually place or adjust this reference point, ensuring complete control over the final stenosis calculation.

4.3.2. Clinical Workflow Integration

The highest translational value of this project lies in embedding the dashboard inside the existing Artificial Intelligence clinical pathway at Hospital de la Santa Creu i Sant Pau [1]. However, before deploying the product for routine commercial use within the hospital environment, it is absolutely critical to resolve all the geometric and technical limitations mentioned earlier. Fortunately, these challenges can be overcome through targeted iterative work on the current pipeline. Once optimized, uniting this quantification tool with other existing projects would generate immense clinical value. Specifically, integrating the automatic coronary segmentations developed by Clapers Colet [10] and the patient prioritization framework by Ferrer Beltran [14] would create a highly robust system. A complete deployment would pass cases that are already segmented and ranked by risk directly into this visualization stack.

In practical terms, a cardiologist reviewing a prioritized list of patients could open the dashboard for a rapid initial assessment. They could easily inspect the global anatomy, review summaries oriented towards CAD-RADS 2.0 categories, and locate zones with a high plaque burden. The doctor can then decide whether the case requires an immediate deep review or if it can be deferred. This workflow preserves the authority of the clinician while drastically reducing time spent on manual navigation. To guide the evolution of this interface, commercial platforms like Cleerly represent an ideal technical ceiling. Analyzing such advanced visualization tools provides an excellent reference for the specific features and interactive elements that cardiologists need to streamline their daily workflow [11].

4.3.3. Expert Clinical Proposals: Reference Modeling and Dashboard Extensions

To directly address the geometric overestimation and rigid reference window limitations discussed in Section 4.2, clinical experts at Hospital de la Santa Creu i Sant Pau proposed moving beyond the fixed 10 mm bilateral window toward an expected healthy vessel model. Rather than trusting adjacent samples that may fall inside bifurcations, tapering zones, or diffuse disease, this approach first characterizes each centerline neighborhood and then estimates what the lumen caliber should be under healthy physiology. The proposed methodology can be summarized as follows:

- **Comprehensive Feature Extraction**
  Clinical experts noted that a robust reference model cannot rely on luminal area alone. For each centerline point, the system should record complementary geometric descriptors such as effective diameter, geodesic distance from the ostium, local curvature, and proximity to bifurcations. Together, these features help the algorithm decide whether a neighborhood is anatomically eligible to serve as healthy reference tissue.

- **Exclusion Of Unreliable Zones**
  Before estimating any expected healthy caliber, the algorithm must explicitly exclude zones that specialists already avoid during manual reading. This includes segments with evident stenosis, bifurcation areas, ostial transitions, severe calcification artifacts, or abrupt non-physiological diameter drops. Formalizing this exclusion step reproduces the clinical habit of staying away from morphologically unstable regions.

- **Robust Reference Curve Fitting**
  Rather than averaging neighboring samples, clinical experts recommended fitting a smooth reference caliber function along each valid branch segment. Methods such as robust spline regression, LOESS, or hierarchical Bayesian smoothing could represent physiological tapering from proximal to distal anatomy. Stenosis would then be computed against the expected area at each exact coordinate:

$$\%AS = 100 \times \left(1 - \frac{A_{\text{observed}}}{A_{\text{expected}}}\right) \tag{4.1}$$

- **Priors Derived From Populations**
  To prevent patient-specific fits from becoming unstable on short branches or noisy distal segments, the expected healthy curve should be constrained by priors learned from anatomically normal coronary cohorts. These priors could be conditioned on territory, segment nomenclature, dominance pattern, sex, body size, and myocardial volume. This reduces overfitting and keeps distal calibers physiologically plausible.

Beyond this mathematical proposal for reference estimation, the evaluation sessions highlighted specific interface improvements that could be implemented much sooner.

Interactive Dashboard Extensions

The previous proposal focuses on improving reference estimation through an expected healthy vessel model, but its full implementation remains future work. Furthermore, several interface enhancements could already mitigate current limitations. Cardiologists requested manual adjustment of proximal and distal reference points, continuous lumen profiles instead of isolated samples, and aggregated stenosis zones to accelerate routine reading. They also noted that clinicians should be able to open the original scan alongside the automated results, so they can check whether a reported stenosis reflects true disease or a geometric artifact. Curved multi-planar reformation and true transverse cross-sections would support this direct comparison with the real anatomy.

Chapter 5

CONCLUSIONS

The primary objective of this bachelor's thesis was to design and develop an automated computational workflow capable of bridging the critical gap between raw coronary segmentations and actionable clinical insights. By directly addressing the operational bottleneck of manual Coronary Computed Tomography Angiography assessment, this project successfully delivered a complete end-to-end pipeline specifically tailored to the clinical ecosystem of Hospital de la Santa Creu i Sant Pau. The resulting framework reliably transforms segmented anatomical data into quantitative stenosis metrics, standardizes the reporting process through automated CAD-RADS 2.0 scoring, and provides an interactive visualization interface for rapid clinical review.

From a technical perspective, the pipeline demonstrates a highly robust integration of complex geometrical algorithms. The automated processing sequence successfully isolates the right and left coronary trees, extracts their mathematical centerlines, and computes orthogonal cross-sectional areas without requiring manual intervention. The mathematical validity of the core quantification logic was strictly verified using synthetic models, confirming that the underlying percentage area stenosis formula operates with high precision in controlled environments. Furthermore, testing across the ASOCA and MACS-18 cohorts proved the system's execution stability and highlighted its capacity to adapt to different levels of anatomical detail.

The culmination of this technical workflow is the interactive clinical dashboard. Designed explicitly to avoid black-box behavior, the interface successfully synchronizes global 3D anatomical renderings with continuous 2D longitudinal metrics. This transparent design empowers cardiologists to visually inspect the exact locations of maximum stenosis and directly verify the proximal and distal reference points used by the algorithm. The pilot evaluation with medical experts confirmed the practical utility of this approach, achieving a positive usability score and demonstrating strong potential to accelerate patient triaging and reduce the cognitive load of diagnostic reading.

However, the transition from synthetic verification to real human anatomy revealed important technical limitations. The current reliance on a rigid reference window causes localized stenosis overestimation in morphologically complex regions, such as severe tapering zones and major bifurcations. Consequently, the current iteration of the pipeline lacks autonomous diagnostic predictive value and must strictly function as a clinical decision-support aid. This outcome directly reinforces the fundamental premise of the project: automated tools are designed to streamline and assist the medical workflow, but the final diagnostic authority must always remain firmly with the specialized clinician.

In conclusion, this project establishes a highly functional, scalable, and explainable foundation for automated coronary artery disease assessment. By outlining a clear path forward toward an expected healthy vessel model, the structural limitations identified during this research can be systematically resolved in future iterations. Ultimately, integrating this quantitative visualization dashboard with the existing automated segmentation and patient prioritization tools represents a significant step toward a more efficient and standardized diagnostic pathway at Hospital de la Santa Creu i Sant Pau [1].

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Appendix A

SUPPORTING MATERIAL

This appendix collects supplementary material referenced from Chapters 2 and 3: VMTK centerline extraction (Section A.1), the SUS questionnaire (Section A.2), extended pipeline distribution charts (Section A.3), synthetic validation visualizations (Section A.4), and Block 4 dashboard screenshots (Section A.5).

A.1. VMTK Centerline Extraction

Block 1 (Section 2.3) invokes `vmtkCenterlines` after automated seed placement on the smoothed lumen surface. This section summarizes how that VMTK function derives centerlines from the triangulated mesh and user-defined source and target seeds. The geometric kernel follows the Vascular Modeling Toolkit as documented by Izzo et al. and Piccinelli et al. [19, 27]. In this thesis, the contribution lies in automated seeding, branch packaging, and downstream integration rather than in reimplementing the centerline solver itself.

Inputs: The function receives the closed 3D surface mesh produced in Phase 2, together with one source point at the ostium and one or more target points at distal endpoints, all snapped onto mesh vertices. Intuitively, the mesh is a triangular approximation of the lumen boundary, and the seeds specify where each centerline must begin and end. These inputs constrain which vessel branches are recovered from the global coronary geometry.

Medial Axis Representation: VMTK first computes a Voronoi diagram from the surface tessellation (the triangular mesh covering the vessel wall). A Voronoi diagram is a geometric structure that divides space according to proximity to the vessel surface. In practical terms, it identifies special interior locations that are as far as possible from the surrounding wall while still remaining inside the lumen. These locations are the Voronoi vertices contained within the vessel.

Each interior Voronoi vertex acts as the center of a maximal inscribed sphere (the largest sphere that can fit at that location without crossing the vessel wall). Collectively, these vertices form a branching network inside the lumen, known as the medial axis. Intuitively, this skeleton is a set of candidate points running through the middle of the artery. Importantly, this step only builds the set of candidate interior points. It does not yet define the final centerline.

Path Extraction: The centerline is obtained by selecting one connected route through the skeleton for each source-to-target pair. VMTK does not use all skeleton points at once. Instead, for each branch, it evaluates the many possible routes that link the ostium seed to the corresponding distal target seed along the network of interior Voronoi vertices. Among all these candidate routes, VMTK selects the one that is most centered overall along the full path, favoring trajectories that remain in the geometric middle of the vessel rather than briefly passing through a locally wide region. This selected path also respects the branch topology through bifurcations and curved segments. Only the points lying on this route are kept as the centerline for that branch.

Outputs And Role In The Pipeline: The selected route is first represented as an ordered but irregularly spaced set of points along the medial axis. VMTK therefore resamples it into a regular sequence of 3D coordinates so that the exported centerline can be stored and processed consistently in later pipeline stages. This step is necessary because skeleton vertices are not uniformly distributed in space, whereas downstream branch packaging and Block 2 quantification require a stable ordered point sequence along each vessel path. At every resampled point, VMTK stores `MaximumInscribedSphereRadius` (the radius of the largest sphere centered on that point that remains inside the lumen surface). Intuitively, this value provides an initial estimate of local vessel width at each centerline station. Block 1 exports these curves as centerline meshes and tabular records. Subsequent topology labeling, branch filtering, and Block 2 cross-sectional quantification operate on this VMTK output rather than on the internal Voronoi representation.

A.2. System Usability Scale Questionnaire

Section 3.4 reports the raw Likert responses in Table 3.5 and the computed scores in Table 3.6. The table rows labeled Question 1–Question 10 correspond to the ten standard SUS items listed below, administered after a guided Block 4 session (Section 2.7.5). Each statement was rated on a five-point Likert scale (1 = strongly disagree, 5 = strongly agree). Item wording follows the standard SUS format [7].

1. I think that I would like to use this coronary viewer frequently in clinical practice.
2. I found the system unnecessarily complex.
3. I thought the Dashboard was easy to use and intuitive.
4. I think that I would need the support of a technical person to be able to use this system autonomously.
5. I found the various functions in this viewer (interactive 3D meshes, profile plots, clinical metrics) were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most medical professionals would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system to analyze artery anatomy and CAD-RADS 2.0 category assignment.
10. I needed to learn a lot of things before I could get going with this system.

Optional open question.

> Do you have any additional comments, suggestions for improvement, or features you would like to see in the future?

Scores were computed using the standard SUS conversion described in Section 2.7.5 and interpreted with the bands in Table 2.8.

A.3. Extended Pipeline Metrics

Section 3.1 summarizes cohort outcomes in Tables 3.1–3.3 (endpoint counts, OISR/CESR, and mean runtimes). The two figures below provide the underlying distributions and grouped comparisons that complement those tables for the same 40-patient ASOCA and MACS-18 cohorts.

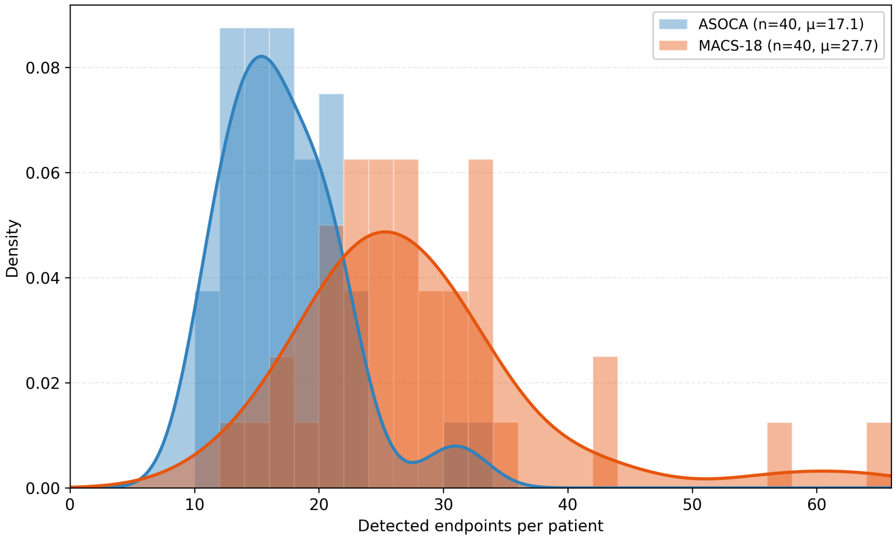

Figure A.1: Per-patient distribution of detected endpoints for ASOCA and MACS-18 cohorts. Extends the summary statistics in Table 3.1.

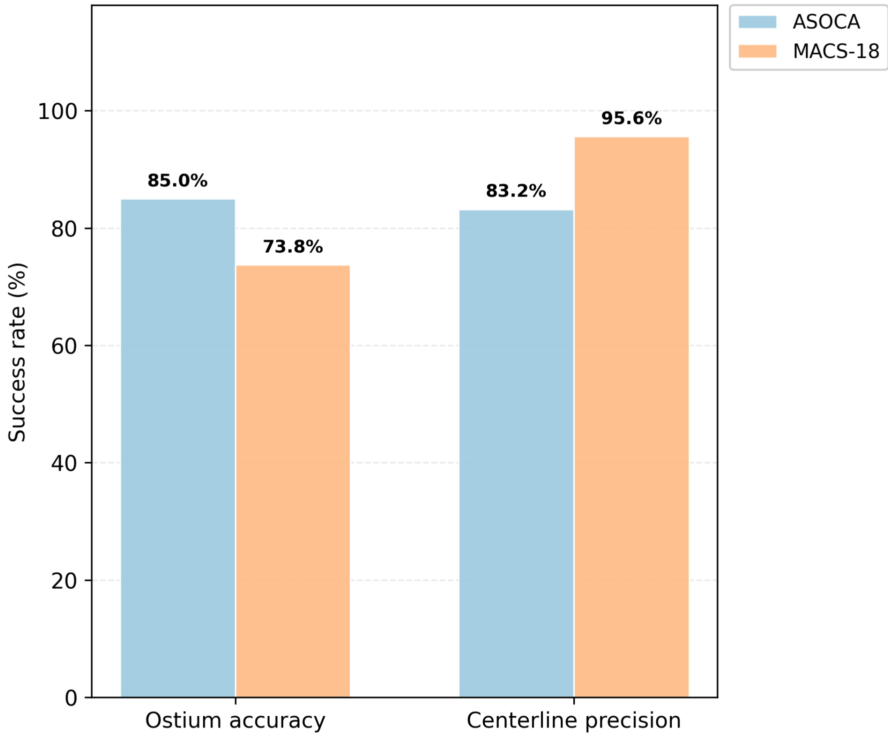

Figure A.2: Grouped comparison of OISR and CESR between ASOCA and MACS-18. Visual counterpart to Table 3.2.

A.4. Synthetic Geometry Profiles

Section 3.2 reports quantitative Block 2 verification in Table 3.4 and shows the reconstructed tubes in Figure 3.2. The figures below provide the extended visual evidence for the same two models: `synthetic_1` (healthy, uniform radius) and `synthetic_2` (diseased, mid-vessel stenosis).

Longitudinal area and %AS profiles along each vessel axis:

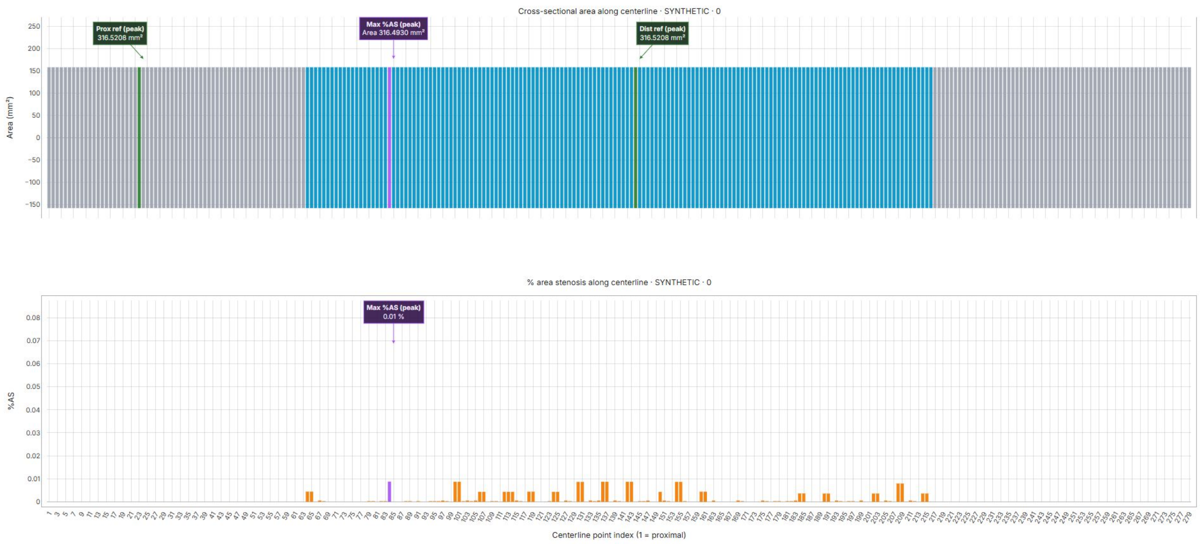

Figure A.3: Longitudinal area and %AS profiles along `synthetic_1`.

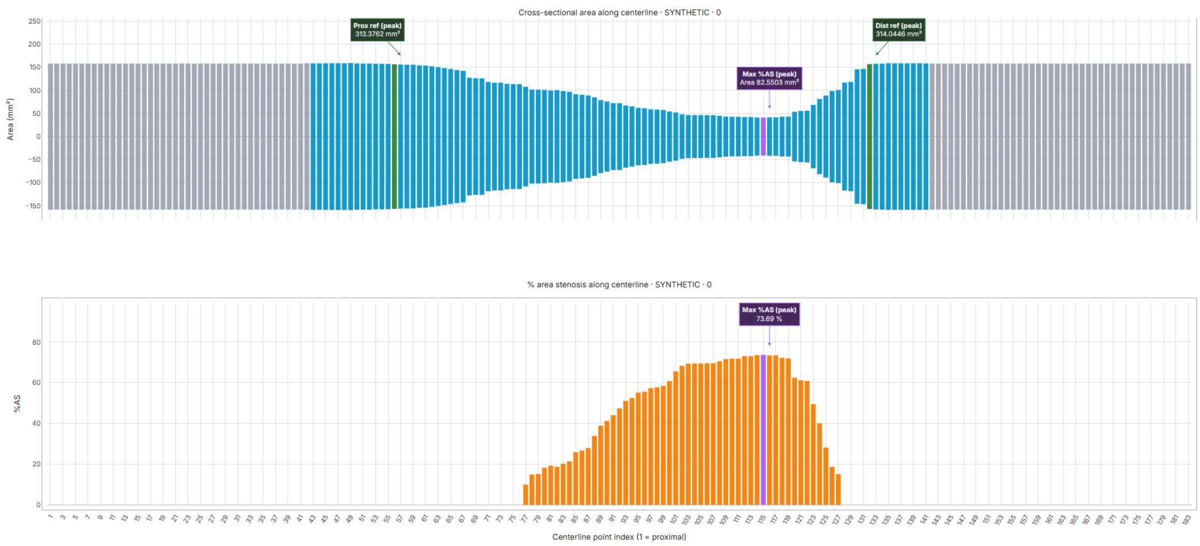

Figure A.4: Longitudinal area and %AS profiles along `synthetic_2`.

Three-dimensional area and %AS colormaps on the reconstructed tube surfaces:

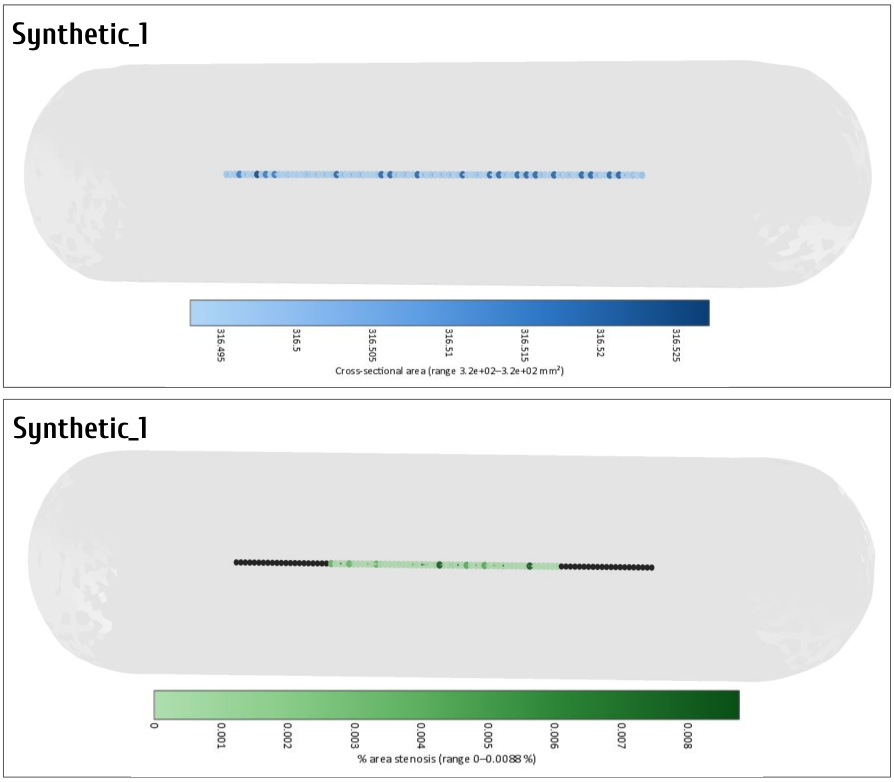

Figure A.5: Three-dimensional area and %AS colormaps for `synthetic_1`.

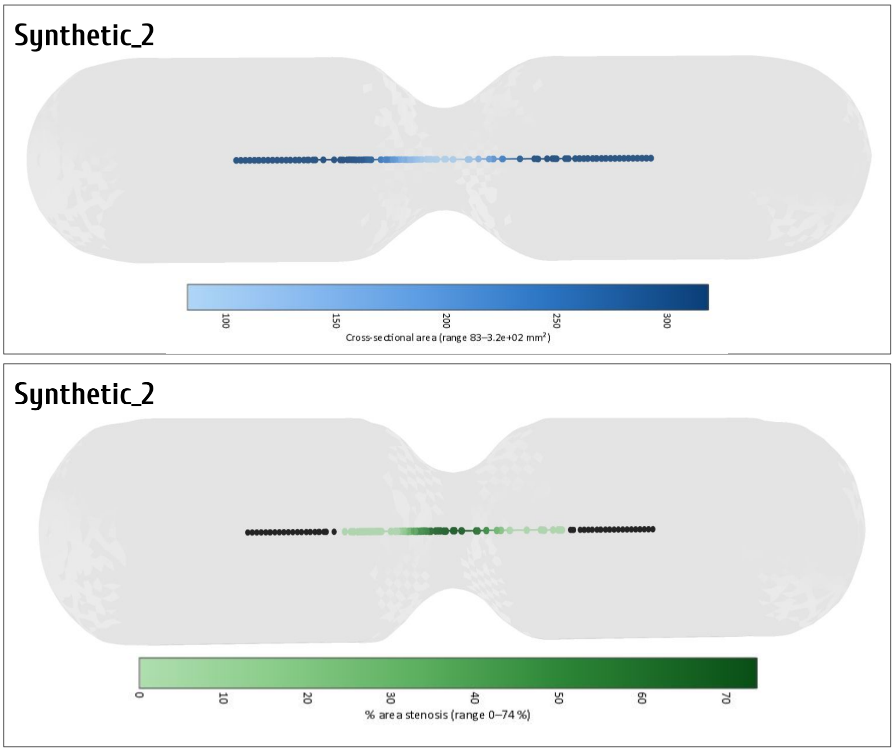

Figure A.6: Three-dimensional area and %AS colormaps for `synthetic_2`.

A.5. Extended Dashboard Interfaces

Section 3.3 describes the Block 4 visualization output produced when validated pipeline results are loaded into the dashboard. The six screenshots below follow the clinical review workflow from the landing view through Modules 1–4, from global anatomy to branch-level metric inspection.

Figure A.7: Block 4 landing view showing the dashboard title, a brief description of the tool, and the patient identifier currently under review.

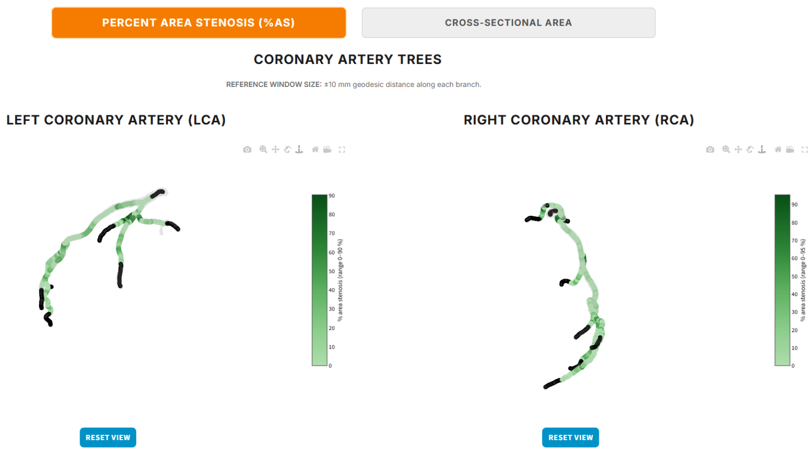

Figure A.8: Module 1: interactive global RCA and LCA three-dimensional renderings with metric colormaps.

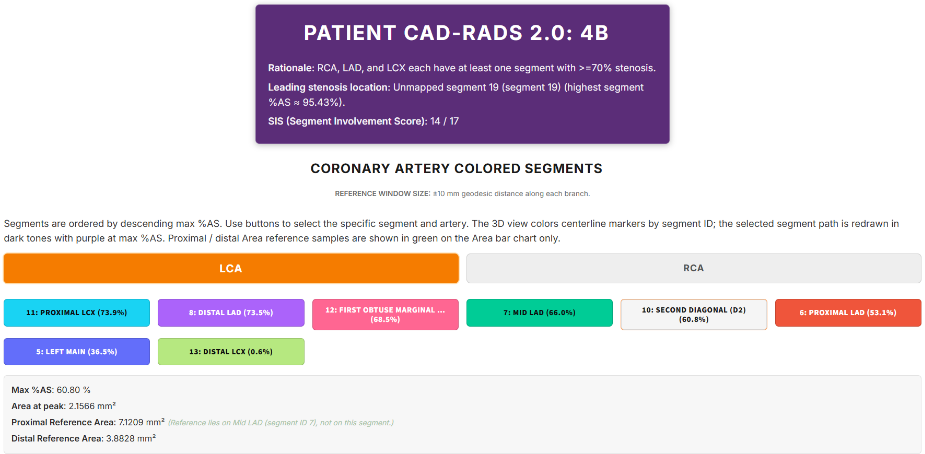

Figure A.9: Modules 2 and 3 (initial view). Module 2 reports the automated patient clinical summary, including the CAD-RADS 2.0 category, segment involvement score (SIS), and territory rationale. Module 3 adds the selected segment label, artery-level context, and interactive segment-selection buttons; the three-dimensional artery view is colored by AHA segment, with each segment color matching its corresponding button. A table summarizes key segment metrics: maximum %AS, area at the peak stenosis, proximal and distal reference areas, and the reference half-window W used for quantification.

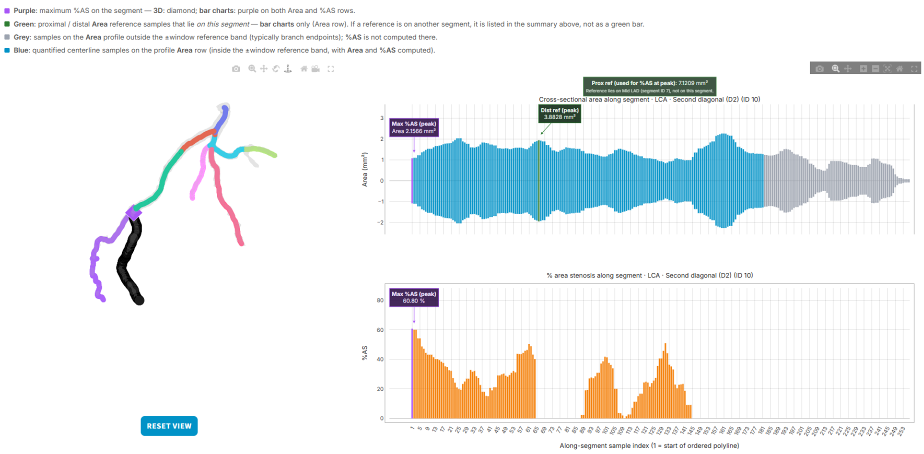

Figure A.10: Module 3: segment-level three-dimensional anatomy with synchronized longitudinal profiles. The artery is rendered with a %AS colormap and annotated in three dimensions with indicator points for the maximum stenosis location within the selected segment and for the proximal and distal reference points. The blue bar plot shows cross-sectional area and the orange bar plot shows %AS along the selected segment; peak stenosis and reference markers are also marked in the profiles. A color legend identifies each visualization component.

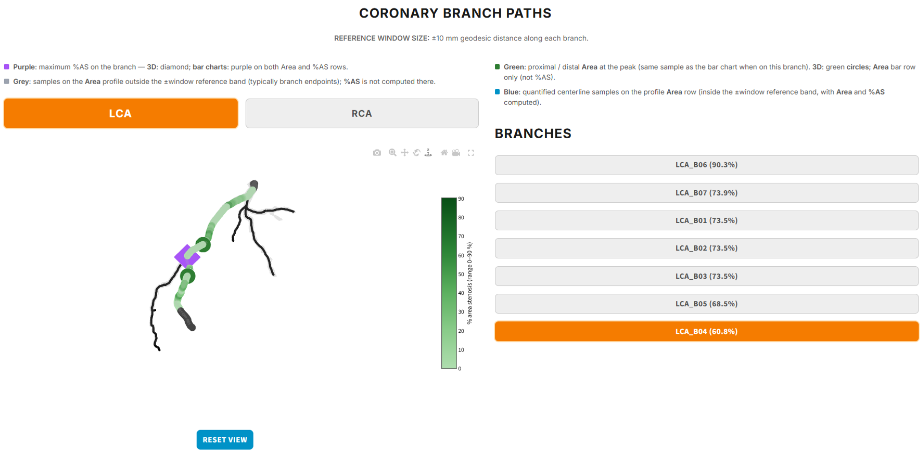

Figure A.11: Module 4: reference half-window W and interactive controls to select the artery and branch under review, together with branch-level three-dimensional path highlighting and longitudinal metric profiles. A color legend identifies each visualization component.

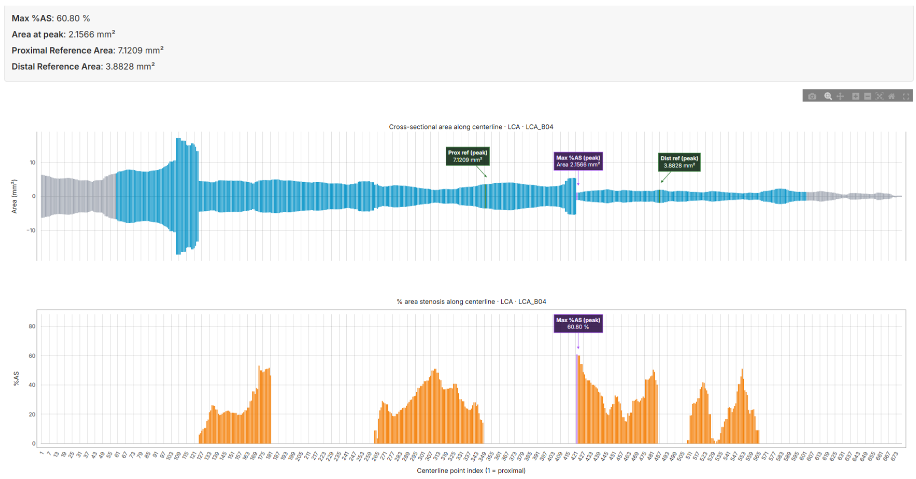

Figure A.12: Module 4 (continued): full-branch area and %AS profiles displayed in proximal-to-distal order, with a summary of branch-level metrics (maximum %AS, area at the peak stenosis, and proximal and distal reference areas).

Appendix B

SOURCE CODE ACCESS

The complete implementation of the methods and experiments presented in this thesis is available at the following repository:

`https://github.com/4dryy/UPF_TFGRepository.git`